How Many Candidates Does a One-Step World Model Need? A Random-Codebook Account of Nearest-of-m Training on Frozen DINO Tokens

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Abstract

One-step stochastic world models avoid the mode averaging of deterministic feature predictors without the tens of network evaluations per step that diffusion and flow matching require. The simplest such objective is nearest-of-m matching (conditional implicit maximum likelihood estimation, IMLE): draw m candidate futures and pull only the one closest to the observed outcome. Recent work conjectures that a small m suffices for feature-space world models and, conversely, that a strictly proper energy score should replace the hard minimum. We test both claims with one-step world models on frozen DINOv2/v3 patch tokens, using benchmarks whose futures can be enumerated. Under mode-like uncertainty, nearest-of-m with m=16 covers the enumerated futures better than 50-step flow matching, one-step flow generators and an energy-score generator at matched parameter counts, with about 30× lower latency than 50-step flow matching, and its rollouts stay on the token manifold where flow rollouts drift. The candidate budget is governed by one quantity: the nearest-of-m objective is the expected distortion of a random codebook of m draws from the generator. This view yields three predictions, which we recorded before the corresponding runs: (i) memorization of the realized outcome in contexts that never repeat, whose strength is governed by the codebook's miss *distortion* $I(m)$ rather than by its miss probability, and which an energy-score generator does not escape by raising m; (ii) a flattening of mode probabilities with m that matches a known optimum within 0.01; and (iii) under-dispersion on continuous high-dimensional outcomes, which bounds the method on real robot video. Per-step matching on self-rollouts cuts long-horizon error by a third on a stochastic Push-T variant, and an ablation separates its manifold benefit from its coverage benefit. Planning success does not improve despite accurate risk estimates, a negative result that we report together with its diagnosis.

1 Introduction

World models that predict the future in the feature space of a frozen vision foundation model are cheap to train and plan with (Zhou et al., 2025; Baldassarre et al., 2025; Assran et al., 2025), but a deterministic predictor returns the conditional mean, which lies between the modes of a multimodal future (Porcher et al., 2026). Stochastic feature-space world models fix this with diffusion or flow matching at tens of network evaluations (NFE) per step (Porcher et al., 2026; Boduljak et al., 2025; Walker et al., 2025), a cost that matters when a planner imagines thousands of futures per decision. Several recent works therefore build one-step stochastic world models (Lu et al., 2026; AgiBot Research Team et al., 2026; Lee et al., 2026a; Park et al., 2026; Hsu et al., 2026).

A simpler one-step objective predates these methods: draw m candidate futures from a noise-conditioned generator, find the one closest to the observed outcome, and pull only that one towards it. This objective is conditional implicit maximum likelihood estimation (IMLE; Li & Malik, 2018; Li et al., 2019); it is also

the min-of-K objective of explorative modeling (XM; Gladstone et al., 2026), whose K is the budget we call m, and a sampled relative of winner-takes-all learning (Guzman-Rivera et al., 2012; Rupprecht et al., 2017). Because every data point is explained by at least one sample, mode collapse is discouraged by the objective itself, and inference is a single forward pass. IMLE world models beat Gaussian ones on proprioceptive state (Aghabozorgi et al., 2026), and Gladstone et al. (2026) conjecture that feature-space world models have few modes, so that a small candidate budget should suffice. In the opposite direction, Makabe et al. (2026) note that a hard minimum over finitely many candidates is not a strictly proper scoring rule and show that an energy-score generator covers a two-branch toy distribution better than min-of-K.

This paper asks a quantitative version of both claims: *how many candidates does a one-step nearest-of-m world model need, when is no budget enough, and is a strictly proper score the better objective?* We answer with a conditional IMLE world model on frozen DINOv2/v3 patch tokens, compared at matched parameter counts against deterministic regression, flow matching with 50 and 2 steps in the FlowWM recipe, two one-step flow generators (improved MeanFlow and Shortcut models) and an energy-score generator, on benchmarks whose futures can be enumerated, so that coverage is measured rather than inferred. One quantity organizes the results: the nearest-of-m loss is, in expectation, the distortion of quantizing the outcome with a *random codebook* of m draws from the generator. This is a sampled version of the known equivalence between winner-takes-all learning and conditional quantization (Rupprecht et al., 2017; Letzelter et al., 2024; Perera et al., 2024), and in the continuous limit it coincides with the Gaussian rate–distortion test channel (Shannon, 1959; Cover & Thomas, 2006). The mathematics is classical; our contribution is to show that this single correspondence predicts, in advance of the runs, where the budget matters and where it cannot help.

Contributions. (1) **A controlled test of the small-budget conjecture.** Under mode-like uncertainty, nearest-of-m with m=16 covers enumerated futures better than 50-step flow matching (FM-50), one-step flow generators and an energy-score generator at matched parameter counts (coverage 0.93 vs. 0.79 and 0.36 on non-repeating contexts; F1 error 7.2 vs. 12.3 px on Bouncing Shapes) and at 30× lower latency than FM-50; on a stochastic Push-T variant its precision advantage is significant and its recall advantage consistent but not significant (Section 6). (2) **Outcome memorization and its incentive.** In contexts that never repeat, a conditional generator can reach zero training loss by memorizing the realized outcome of each training context, keyed on its appearance. We measure this as the share of samples on the realized mode of training contexts, show that it is governed by the codebook's miss *distortion* $I(m)$ rather than by the miss probability $(1-1/K)^m$ (fit RMSE 0.031 vs. 0.068 over $K \in \{2,4,8,16\}$), confirm seven of nine predictions recorded before the K=2, 16 runs, and explain why the strictly proper energy score memorizes at every m while nearest-of-m with m=16 does not (Section 7). (3) **Where no budget suffices.** Mode probabilities flatten with m, matching the optimum of Bennett (2026) within 0.01, and the spread on continuous high-dimensional outcomes shrinks to $e_m/\sqrt{d}$ of the true spread, the form of the Gaussian rate–distortion test channel; on BAIR the samples spread along only one to three context-specific directions, and m=64 does not close the gap to multi-step flow matching on distribution metrics (Section 8). (4) **Rollouts and planning.** nearest-of-m rollouts stay as close to the token manifold as deterministic regression, while flow rollouts drift by up to 4× at h=32 regardless of the number of integration steps; per-step matching on self-generated contexts (in the lineage of Venkatraman et al., 2015; Zhang et al., 2025) cuts long-horizon recall error by a third on stochastic Push-T, and an ablation separates its manifold benefit from its coverage benefit; planning with IMLE-WM is 50× faster than with FM-50 at equal success, but success does not improve even where the model's risk estimates are accurate (Section 9).

Negative results are reported in Appendix G. The benchmarks are synthetic or simulated except for BAIR, the models are small (8M and 21M parameters), and the flow baselines are far from the scale of Porcher et al. (2026); Section 10 states these limits.

2 Related Work

Min-of-m objectives. IMLE matches each data point to its nearest generated sample (Li & Malik, 2018); conditional IMLE ignores its noise input when each condition has a single target (Li et al., 2019; 2020), and later variants reject samples that are already close, anneal a per-sample radius, split the noise hierarchically,

or add a robust loss (Vashist et al., 2024; Aghabozorgi et al., 2023; Peng et al., 2022; Vashist & Li, 2026). IMLE has been used as a world model on state vectors (Aghabozorgi et al., 2026), as a trajectory generator for model-predictive control (Lee et al., 2026b) and as an action head (Rana et al., 2025; Hosseinkhani et al., 2026). XM (Gladstone et al., 2026) studies the min-of-K objective as a pretraining axis, includes a one-step world model on Maze2D, and finds that a larger K reduces overfitting on video, which it attributes to expressivity. Conditional min-of-N objectives go back to Fan et al. (2017) and Bhattacharyya et al. (2018), and multi-hypothesis feature forecasting to Vondrick et al. (2016). Winner-takes-all and multiple-choice learning (Guzman-Rivera et al., 2012; Lee et al., 2016; Rupprecht et al., 2017; Makansi et al., 2019; Seo et al., 2020) learn K heads; their optima are conditional quantizers (Letzelter et al., 2024; Delattre et al., 2025; Cortés et al., 2025), their training follows a rate–distortion curve (Perera et al., 2024), and their hard assignment distorts mode weights (Wu et al., 2026). Bennett (2026) derives the optimal mode weights of min-of-K, which we confirm; Makabe et al. (2026) and Branch-JEPA (Song et al., 2026) compare hard assignment with energy-score training.

World models in feature space. DINO-WM, DINO-world and V-JEPA 2 predict frozen features deterministically (Zhou et al., 2025; Baldassarre et al., 2025; Assran et al., 2025); FlowWM (Porcher et al., 2026) runs flow matching in DINOv3 feature space and introduces the Bouncing Shapes benchmark that we adopt; VFMF and Walker et al. (2025) use diffusion on compressed or frozen features. Recent one-step stochastic world models (Lu et al., 2026; AgiBot Research Team et al., 2026; Lee et al., 2026a; Park et al., 2026; Hsu et al., 2026) do not measure mode coverage on enumerable futures, and Lee et al. (2026a) report no planning gain on Push-T. That generated latents leave the manifold of clean encoder features is a known problem (Zheng et al., 2025; Boduljak et al., 2025; Karypidis et al., 2026); we measure it for each objective.

Memorization and rollout training. Conditional diffusion models memorize more when the condition is more informative (Gu et al., 2025; Kim & Lee, 2026); data copying is detected as local over-representation of training points (Meehan et al., 2020; Somepalli et al., 2023; Carlini et al., 2023); conditional GANs that ignore their noise input exhibit a related collapse (Yang et al., 2019; Mao et al., 2019); and with unlimited capacity any sample-based objective memorizes (Li & Malik, 2018). Our measurement labels outcomes by mode and compares training contexts with held-out contexts. Training on self-generated context is the classical remedy for compounding error (Ross et al., 2011; Venkatraman et al., 2015; Bengio et al., 2015; Lamb et al., 2016; Hafner et al., 2019; Chen et al., 2024; Huang et al., 2025; Gao et al., 2026); closest-among-top-K fine-tuning (Zhang et al., 2025) rolls a tokenized policy out on the candidate nearest the ground truth, and our per-step matching is its continuous-feature analogue. Terver et al. (2025) and Cheung & He (2026) find that multi-step rollout losses and one-step fidelity are poor predictors of planning success; our planning results agree.

3 Nearest-of-m World Models on Frozen DINO Tokens

Setting. A frozen encoder maps each observation to patch tokens $s_t \in \mathbb{R}^{N\times D}$ with $N = 256$ tokens on a 16×16 grid and $D = 384$ channels: DINOv2 ViT-S/14 at 224 px (Oquab et al., 2024) for the bounce, push, Push-T and BAIR benchmarks, and FlowWM's DINOv3 feature definition (Siméoni et al., 2025; Porcher et al., 2026) for Bouncing Shapes. Tokens are precomputed and standardized per token. The model takes a context of three token maps c_t, an action a_t and a noise vector $z \in \mathbb{R}^{256}$, and predicts the next map as a residual $\hat{s}_{t+1} = s_t + g_\theta(c_t, a_t, z)$; rollouts feed $\hat{s}_{t+1}$ back into the context, at one network evaluation per step.

Generator. g_θ is a ConvNeXt stack (Liu et al., 2022) on the token grid. The context maps are concatenated along channels; the action is injected by feature-wise linear modulation (FiLM); the noise is injected by adaptive layer normalization through a four-layer mapping network, as in ROMS-IMLE (Vashist & Li, 2026); and the output head is a per-token MLP of width 1024, the "wide head" that FlowWM found necessary in feature space, which we attach to every method so that parameter counts match within ±4%. We use two sizes: width 256 with 8 blocks (8.0–8.2M across methods) for the synthetic benchmarks and width 384 with 12 blocks (20.6M) for Push-T and BAIR. The noise enters only the last three blocks (four for the larger model), so the trunk is evaluated once per transition and only the tail once per candidate; this late fusion halves training time at equal coverage (Appendix D).

Objective. For each transition we draw m noise vectors (the *candidate budget*, or budget for short), generate m candidates without gradient tracking, select the one nearest to the realized outcome in token space,

$$j^\star = \arg\min_{j\in[m]} \left\| s_{t+1} - \hat{s}^{(j)} \right\|^2, \tag{1}$$

and regenerate the selected candidate with gradient tracking, training it with a Geman–McClure loss (Geman & McClure, 1987) on its root-mean-square distance d: $\rho(d) = 2u^2/(u^2+4)$, where $u = d/(2c)$ and c is the running median of the matching distance. The matching is recomputed for every batch with a per-transition candidate pool (as in WIMLE; Aghabozorgi et al., 2026), in bfloat16, without gradient tracking, and in a separate mixed-precision region (Appendix D describes a pitfall of sharing one). The robust loss reproduced the gain reported for ROMS-IMLE in our pilot study but matters little in the final recipe (Appendix D); the budget m is the variable this paper studies. We call the resulting model IMLE-WM. Every 2,000 steps, a *memorization monitor* draws 16 samples for each of 200 training contexts and 200 held-out contexts whose next transition is stochastic, and records the share of samples decoded onto the mode that was realized in the data. This share is $1/K$ in expectation on held-out contexts, and it exceeds $1/K$ on training contexts only if the model has learned which mode each training context took.

Per-step matching on self-rollouts. Matching whole H-step candidate trajectories fails: the number of trajectory modes grows like K^H, so the selected candidate is the one that is close to all of them on average (Appendix G). We instead match one step at a time along the model's own rollout. Starting from a real context, at each step $h = 1, \ldots, H$ we draw m fresh noise vectors, select the candidate nearest to the true frame s_{t+h} *given the self-predicted context*, train that step, and feed the selected prediction back (detached) as the next context; with H=4 this costs mH forward passes without gradient tracking and H with it. Section 9 compares two alternatives for the fed-back state, a random candidate and the true frame. Selecting the candidate nearest the ground truth is the device of closest-among-top-K rollouts (Zhang et al., 2025); in nearest-of-m training this selection is the argmin of the loss itself.

Inference, planning and baselines. One predicted step is one forward pass, and m_p futures for a state are obtained by batching m_p noise vectors. Planning uses cross-entropy-method model-predictive control (CEM-MPC; Rubinstein, 1999; Zhou et al., 2025), scoring each candidate action sequence by the mean, over m_p imagined futures, of a distance to the goal; a variance penalty lowered the success rate even with the true simulator as the model (Appendix G). All baselines share the backbone, wide head, context, residual parameterization and training budget unless stated otherwise: *regression* (squared error); *FM-50*, flow matching (Lipman et al., 2023) in the FlowWM recipe (time shift $\alpha = 13$, 50 Euler steps), and *FM-2*, the same model sampled with 2 steps; *iMF* (Geng et al., 2025a;b), ported from the official code and trained for 3× as many steps; *Shortcut* (Frans et al., 2025; Hafner et al., 2025), trained to the wall-clock budget of IMLE-WM with m=16; and the *energy score* (Gneiting & Raftery, 2007) as used by Makabe et al. (2026), i.e. the IMLE-WM generator trained with the unbiased m-sample estimator of $\mathbb{E}\|Y - x\|^\beta - \frac{1}{2}\mathbb{E}\|Y - Y'\|^\beta$ ($\beta = 1$, whole-map distance, $m \in \{8, 16\}$), plus a per-token variant (Appendix D).

4 A Random-Codebook Account of Nearest-of-m Training

Fix a context and write p for the true outcome distribution and q for the generator's distribution. Since the candidates are drawn independently from q, the population objective behind Equation 1 is

$$L(q) = \mathbb{E}_{x\sim p}\, \mathbb{E}_{Y_1,\ldots,Y_m\sim q} \min_{j\in[m]} \|Y_j - x\|^2, \tag{2}$$

the expected distortion of quantizing x with a random codebook of m i.i.d. draws from q. Winner-takes-all learning is the deterministic version of this problem, whose minimizers form centroidal Voronoi tessellations of the conditional distribution (Rupprecht et al., 2017; Letzelter et al., 2024); for a Gaussian source, the random version is Shannon's random-coding construction at a rate of $\ln m/d$ nats per dimension (Shannon, 1959). The consequences below are elementary (proofs in Appendix A); their value lies in predictions that we wrote down before the corresponding runs.

Outcome memorization in non-repeating contexts. Let the outcome of a context take one of K modes $\mu_1, \ldots, \mu_K$. In data whose contexts never repeat, each training context c_i is seen with a single realized mode

μ_{k_i}, and two solutions compete for the empirical loss: the *memorizing* solution $q(\cdot \mid c_i) = \delta_{\mu_{k_i}}$, which attains zero loss at the cost of storing k_i for every training context (and therefore requires a feature of c_i that identifies it), and the *generalizing* solution $q = p$, which attains

$$I(m) = \mathbb{E}_{k\sim p}\, \mathbb{E}_{S_1,\dots,S_m\sim p}\, \min_j \|\mu_{S_j} - \mu_k\|^2, \tag{3}$$

the distortion of a random codebook of m mode draws.

Proposition 1 (memorization incentive)**.** *With non-repeating contexts, the empirical nearest-of-m loss of the memorizing solution is lower than that of the generalizing solution by exactly $I(m)$. For equiprobable modes with pairwise distances in $[\delta, \Delta]$, $(1-1/K)^m\delta^2 \le I(m) \le (1-1/K)^m\Delta^2$, so the incentive vanishes exponentially once m exceeds K; if the modes densely fill a curve of length ℓ with $K \gg m$, then $I(m) = \Theta(\ell^2/m^2)$, independently of K. For squared-error regression the corresponding incentive is* $\operatorname{tr}\operatorname{Cov}_p$*, which no budget reduces.*

The incentive is a *distortion*, not a probability: a missed mode is weighted by the distance to the nearest drawn mode, and whether K enters at all depends on the geometry. On our bounce benchmark the K modes are equiangular directions on a semicircle, so for $m \lesssim K$ the distortion is nearly the same for every K (Appendix B computes it exactly); the rule of thumb "$m \approx 2K$" that the two values $K \in \{4, 8\}$ alone would suggest is a coincidence, and Section 7 rejects it.

High-dimensional continuous outcomes.

Proposition 2 (isotropic shrinkage)**.** *Let $p = \mathcal{N}(\mu, \sigma^2 I_d)$ and $q = \mathcal{N}(\mu, s^2 I_d)$. For large d, $L(q) \approx d\sigma^2 + ds^2 - 2s\sigma\sqrt{d}\, e_m$ with $e_m = \mathbb{E}\max_{j\le m}\xi_j$, $\xi_j \sim \mathcal{N}(0,1)$, so the optimal spread is $s^\star/\sigma = e_m/\sqrt{d}$ ($e_4 = 1.03$, $e_{16} = 1.77$, $e_{64} = 2.34$; $e_m \approx \sqrt{2\ln m}$ for large m).*

This is the first-order form of the reproduction variance $\sigma^2(1 - m^{-2/d})$ of the Gaussian rate–distortion test channel (Cover & Thomas, 2006), and the correspondence is exact in construction because IMLE draws its codebook i.i.d. from the reproduction distribution. It gives a dichotomy between $\ln m$ and the effective dimension d_{eff} of the conditional outcome: a few discrete modes are covered once $\ln m > \ln K$ (the regime of Proposition 1, in which an i.i.d. min-of-N loss over-disperses low-dimensional densities; Thiede & Brahma, 2019), whereas when $d_{\text{eff}} \gg e_m^2$ the samples contract towards the mean and their spread grows only like $\sqrt{\ln m}$.

Flattening of mode probabilities. Even when every mode is covered, the population optimum for equidistant modes satisfies $p_k(1-q_k)^{m-1} = \text{const}$ (Bennett, 2026, Theorem 7): at m=1 it collapses onto the most frequent mode, at K=2 and m=2 it is calibrated, and for large m the differences $q_k - q_l \approx \ln(p_k/p_l)/(m-1)$ vanish, so rare modes are over-represented; for equiprobable modes the optimum is exactly $q = p$ for every $m \ge 2$ (Bennett, 2026, Theorem 4). The budget therefore trades coverage against calibration, which benchmarks with equiprobable modes cannot reveal (Section 8).

The energy score's incentive does not depend on m. Makabe et al. (2026) prefer the energy score $\mathrm{ES}_\beta(q, x) = \mathbb{E}_{Y\sim q}\|Y - x\|^\beta - \frac{1}{2}\mathbb{E}_{Y,Y'\sim q}\|Y - Y'\|^\beta$, $\beta \in (0, 2)$, because it is strictly proper. Strict propriety is a population property; in the finite-sample regime above, the energy score has a further consequence.

Proposition 3 (memorization incentive of the energy score)**.** *(a) $\mathrm{ES}_\beta(q, x)$ is half the energy distance between q and δ_x (Dawid, 2007; Gneiting & Raftery, 2007), so with non-repeating contexts the unique minimizer of the empirical risk is the memorizing solution, for every $m \ge 2$. (b) The generalizing solution attains $\frac{1}{2}\mathbb{E}_{Y,Y'\sim p}\|Y - Y'\|^\beta$, so the incentive is half the expected outcome spread and does not depend on m (m changes only the variance of the estimator). (c) For $\beta = 2$ the score reduces to squared-error regression. (d) A per-token score averaged over the token grid inherits (a) and (b).*

Normalizing each incentive by the incentive of its fit term alone ($\mathbb{E}\|Y - x\|^\beta$ for the energy score, $I(1)$ for nearest-of-m) gives a unit-free comparison: regression and the energy score sit at 1/2 for every m, whereas nearest-of-m sits at $I(m)/I(1)$, which is 0.52, 0.21, 0.054 and 0.005 at $m = 2, 4, 8, 16$ for K=4 on the bounce geometry. Parts (a)–(c) are classical (Gneiting & Raftery, 2007; Dawid, 2007), and Li & Malik (2018) note the unconditional case of (a); the comparison and its empirical test (Section 7) are new. A second difference

Table 1: Benchmarks. "Contexts repeat" means the same context appears in the training data with several outcomes, which removes the possibility of memorizing the realized outcome. Details in Appendix C.

Name	Source	Stochasticity	Repeat	Actions	Model
A1 Bouncing Shapes	FlowWM generator	normal bounce vs. reversal per collision	yes	none	8M (joint 27–29M)
A2-div / A2-fix bounce	ours	$K \in \{2, 4, 8, 16\}$ directions at a wall; asymmetric K=2	no / partly	none	8M
Push (branch risk)	ours	$\pm 20°$ (2 modes), ice $\{-60°, 0, 60°\}$ (3), absorbing pit	no	2-D	8M
B stochastic Push-T	DINO-WM data, re-simulated	hidden center of gravity, 4 modes	no	10-D	21M
BAIR robot pushing	Ebert et al. (2017)	continuous, high-dimensional	no	4-D	21M

concerns the *direction* of the spread: with an unknown bias of length a and an isotropic spread s in d dimensions, the energy score is minimized at $s = a$ independently of d (a special case of Proposition 4.1 of De Bortoli et al., 2025), whereas Proposition 2 gives $s \approx a\, e_m/\sqrt{d}$. The energy score thus calibrates the size of the spread but is insensitive to its direction in high dimensions (Chakraborty & Zhang, 2021; Pinson & Tastu, 2013; Scheuerer & Hamill, 2015; Marcotte et al., 2023), so an isotropic spread leaves the token manifold, whereas nearest-of-m under-disperses but stays on it (an interpretation; Appendix A).

The predictions these propositions make, and where each is tested, are listed in Appendix B.

5 Experimental Setup

Benchmarks (Table 1; Appendix C). *A1*: the public FlowWM generator, used unchanged (Porcher et al., 2026): a square and a circle that, on hitting a wall, bounce normally or reverse their velocity with probability one half; 16,384 training clips (one eighth of FlowWM's) from 5,324 initial conditions with 3.08 futures each on average, so contexts repeat; FlowWM's DINOv3 features, protocols and heatmap readout. *A2*: a ball moving in a 64×64 frame whose direction after hitting a wall is drawn uniformly from K equiangular directions (about 28% of transitions are stochastic; no actions). A2-div redraws the background color, ball color, radius, speed and two distractor rectangles for every episode, so the nearest training context of a test context is 1.9× farther away than the typical nearest neighbor within the training set and every training context carries an identifying key; A2-fix keeps the appearance fixed (0.35×). Each setting has 40k training frames and 400 held-out wall contexts; an asymmetric K=2 variant has mode probabilities $(0.8, 0.2)$. *Push*: the same renderer with a 2-D push action deflected by $\pm 20°$ in the open region (two modes) and by $\{-60°, 0, 60°\}$ on an ice patch beside an absorbing pit (three modes); in the *branch-risk task* the path to the goal runs along the pit, so one of the three ice modes leads into it. *B*: the Push-T action sequences of DINO-WM (Zhou et al., 2025) replayed in the simulator with the T block's center of gravity hidden at one of four positions per episode (mass and friction have no effect under quasi-static pushing); 2,400 training episodes; a multi-future test set of 128 pre-contact starting states × 64 futures × 16 steps. *BAIR*: 64×64 frames with 4-D actions; 4,096 training and 256 evaluation sequences with 16 samples each.

Metrics. *Coverage* and *precision*: 16 samples per held-out stochastic context are decoded by a position readout (test error 0.13 px; Appendix E) and assigned to the nearest enumerated mode within a tolerance; coverage is the fraction of modes hit, and precision is the fraction of samples that land on a mode (an ideal 16-sample sampler reaches coverage $1 - (1 - 1/K)^{16}$). *Memorization index*: the same share, on the realized mode of 200 training contexts (chance $1/K$; *excess memorization* is $(\text{index} - 1/K)/(1 - 1/K)$). *kNN precision/recall* (Kynkäänniemi et al., 2019) on PCA-reduced tokens; *deterministic-transition error*: the readout error on non-stochastic transitions. *FlowWM precision/recall error* (A1, B): with twice as many rollouts as true futures per starting state, the mean distance from each rollout to the nearest true future (precision), from each true future to the nearest rollout (recall), and their harmonic mean (F1); L1 over the shape centers on A1, L2 over the eight T vertices on B. *Manifold distance*: the distance from a predicted token map to the nearest real one, divided by the median nearest-neighbor distance among real maps (1 = on the data). *Best-of-n*: the readout (or token) error of the sample nearest the truth; *spread*: the mean pairwise distance among the samples. On BAIR we also report token RMS errors, the spread-to-error ratio, and the Fréchet distance and kNN recall of pooled tokens. *Latency* is measured per predicted step at batch size 1; *planning* reports success rate, pit rate and wall-clock time per MPC step.

Table 2: A2-div (K=4, randomized appearance), 8M models, 12k steps, three seeds (energy score m=16: two). Ideal-sampler coverage 0.99; memorization chance 0.25; latency per step at batch 1 on a shared RTX A5000.

Method	NFE	Coverage ↑	Precision ↑	kNN recall ↑	Det. err. (px) ↓	Memo. index	ms/step	Train (h)
Regression	1	0.15 ± 0.01	0.61 ± 0.02	0.00	1.09 ± 0.03	0.99 ± 0.01	4.4	0.3
FM-50 (FlowWM recipe)	50	0.79 ± 0.01	0.51 ± 0.00	0.91 ± 0.00	2.24 ± 0.02	0.29 ± 0.01	107	0.4
FM-2 (same model)	2	0.60 ± 0.00	0.46 ± 0.01	0.56 ± 0.01	1.72 ± 0.01	0.29 ± 0.01	6.5	0.4
iMF (3× steps)	1	0.56 ± 0.01	0.23 ± 0.02	0.92 ± 0.03	14.5 ± 0.7	0.25 ± 0.01	3.4	3.2
Shortcut (2× steps)	1	0.66 ± 0.00	0.42 ± 0.01	0.77 ± 0.01	3.60 ± 0.07	0.27 ± 0.01	4.3	0.7
Energy score, m=8	1	0.36 ± 0.02	0.61 ± 0.02	0.77 ± 0.06	1.06 ± 0.04	0.87 ± 0.06	3.1	1.2
Energy score, m=16	1	0.28 ± 0.01	0.61 ± 0.01	0.60 ± 0.02	1.06 ± 0.03	0.97 ± 0.01	3.3	2.2
IMLE-WM, m=4	1	0.46 ± 0.01	0.68 ± 0.01	0.65 ± 0.01	1.14 ± 0.07	0.75 ± 0.01	3.7	0.6
IMLE-WM, m=16	1	0.93 ± 0.01	0.82 ± 0.01	0.94 ± 0.01	1.55 ± 0.05	0.29 ± 0.00	3.3	1.1

Table 3: Bouncing Shapes F1 error in px (frame 256 px), three seeds. Autoregressive: 3-frame context, 16 one-step predictions, 8M models. Joint: 16 frames predicted at once from 16 context frames, 27–29M models. A last-frame-copy baseline scores 21.7 in every column; iMF, Shortcut and the energy score reach 16.8, 12.1 and 8.9 (autoregressive, probe; Appendix E). Porcher et al. (2026) report 4.31 for FlowWM and 17.1 for their deterministic baseline under their heatmap readout with eight times our training data.

		Autoregressive		Joint 16→16	
Method	NFE	probe	heatmap	probe	heatmap
Regression	1	8.09 ± 0.15	8.41 ± 0.15	5.14 ± 0.01	22.48 ± 0.36
FM-50	50	12.28 ± 0.13	14.11 ± 0.20	3.86 ± 0.02	12.59 ± 0.14
FM-2	2	9.95 ± 0.19	12.49 ± 0.38	3.83 ± 0.02	14.80 ± 0.17
IMLE-WM, m=4	1	6.61 ± 0.18	6.83 ± 0.16	3.30 ± 0.02	11.73 ± 0.31
IMLE-WM, m=16	1	7.21 ± 0.10	7.64 ± 0.18	3.32 ± 0.07	9.04 ± 0.06

Protocol. All models use AdamW (learning rate 2×10^{-4}, $\beta_2 = 0.9$), batch size 128, an exponential moving average of the weights, and 12k training steps unless stated otherwise (8k for the budget grid, the asymmetric variant and per-step matching; 36k for iMF; 24k for Shortcut); parameter counts match within ±4%. Main comparisons use three seeds, grids and follow-up runs two, and ablations one; uncertainty is quantified with bootstrap intervals over contexts and, for Push-T, a hierarchical bootstrap over seeds and starting states (Appendix H). All experiments ran on two RTX A5000 GPUs, about 300 GPU-hours in total. The predictions for the K=2, 16 grid, the energy-score follow-up runs and the fed-back-state ablation were written down before those runs and are reproduced in the appendix, including the ones that missed.

6 Results I: Does a Small Candidate Budget Suffice?

Non-repeating contexts (A2-div). Table 2 is the main comparison (Figure 3, Appendix F). IMLE-WM with m=16 covers 0.93 of the enumerated modes, against an ideal-sampler ceiling of 0.99, with precision 0.82: its samples sit on the modes. FM-50 reaches a coverage of 0.79 with precision 0.51, scattering its samples between the modes; FM-2, iMF and Shortcut are below it on both measures. The energy-score generator has the *lowest* coverage of all stochastic methods (0.36 at m=8, 0.28 at m=16) and, like regression, has memorized the training contexts (memorization index 0.87–0.97), while IMLE-WM with m=4 lies in between (coverage 0.46, index 0.75); Section 7 analyzes this column. The cost of m=16 is a larger error on deterministic transitions (1.55 vs. 1.14 px for m=4) and 1.7× the training time of m=4; the inference cost is independent of m, at 3.3 ms per step against 107 ms for FM-50.

Bouncing Shapes (A1). Table 3 covers two prediction protocols (autoregressive one-step rollouts, and FlowWM's joint prediction of 16 future frames from 16 context frames) and two readouts (our probe and FlowWM's heatmap readout, which we ported). In all four combinations IMLE-WM has the lowest F1 error, by at least ten seed standard deviations: 6.6 vs. 12.3 px for FM-50 in the autoregressive protocol, and 9.0 vs. 12.6 px under FlowWM's readout in FlowWM's own protocol. Contexts repeat here, so there is nothing to memorize and m=4 suffices. The readout matters: the deterministic joint model moves from 5.1

Table 4: Stochastic Push-T multi-future test (128 pre-contact starts × 64 futures × 16 steps): precision / recall error over the eight T vertices in px (frame 512 px), 21M, three seeds (iMF two). GT spread: mean pairwise distance among the true futures.

		h=8		h=16	
Method	NFE	precision	recall	precision	recall
GT spread (reference)		30.8		48.3	
Regression	1	24.8 ± 1.1	39.6 ± 3.4	40.2 ± 3.0	66.4 ± 8.2
FM-50	50	27.1 ± 1.6	24.3 ± 1.8	40.7 ± 1.1	42.0 ± 3.6
FM-2	2	24.2 ± 1.7	23.3 ± 1.7	39.7 ± 1.4	41.5 ± 3.8
iMF	1	37.2 ± 2.6	33.5 ± 3.2	55.1 ± 3.7	52.4 ± 3.4
Shortcut	1	28.0 ± 1.0	24.2 ± 0.4	47.3 ± 1.9	45.9 ± 1.1
IMLE-WM, m=4	1	18.7 ± 0.4	29.2 ± 3.4	33.8 ± 3.2	49.1 ± 3.5
IMLE-WM, m=16	1	21.8 ± 1.8	22.1 ± 2.3	34.1 ± 0.7	37.8 ± 2.3

to 22.5 px when the readout changes, which accounts for most of the gap to the 17.1 px that Porcher et al. (2026) report for their deterministic baseline, whereas the autoregressive models barely move because they emit a clean feature map at every step (Section 9). Appendix F reports the baseline fairness checks (longer training preserves the ranking, iMF saturates after 24k steps, Shortcut is trained to IMLE-WM's wall-clock budget, parameter counts match within ±4%). The caveat is scale: our joint FM-50 reaches 12.6 px under FlowWM's readout, where the FlowWM paper reports 4.31 px with eight times the data, so our claims are confined to matched small-scale compute.

Stochastic Push-T (B). On the Push-T variant with a hidden center of gravity, IMLE-WM with m=16 has the lowest recall error at both horizons and a precision error 6.6 px below that of FM-50 at h=16 (Table 4). Under a hierarchical bootstrap over seeds and starting states, the precision advantage over FM-50, FM-2 and Shortcut is significant (at h=16: -6.6 $[-8.5, -4.7]$, -5.6 $[-7.7, -3.2]$ and -13.2 $[-16.3, -10.2]$ px), whereas the recall advantage, although consistent across the 128 starting states, has an interval that includes zero against the flow baselines (-4.2 $[-8.8, +1.1]$ vs. FM-50; Appendix F). All models mispredict the block's displacement at first contact by about 12 of 16 px, an error that three times more training does not shrink and that masks differences between objectives. IMLE-WM with m=4 shows the collapse analyzed in Section 7: the best precision but the worst recall among the stochastic models, and a rollout spread of 9 px against 22 px for m=16.

7 Results II: Outcome Memorization and the Candidate Budget

The phenomenon. Regression reaches a memorization index of 0.99 on A2-div: it learns which mode every training context took. IMLE-WM with m=4 climbs to 0.75 while its held-out coverage falls from 0.77 at 4k steps to 0.46 at 12k steps (Figure 4, Appendix F); on held-out contexts its index stays at chance, so this is memorization of the realized outcome of each training context, keyed on the context's appearance, rather than a failure to generalize. Training the m=4 model twice as long, or making it 2.7× larger, raises the index further and lowers coverage (Appendix F), so the collapse is not under-training. On A2-fix, where the identifying key is absent, the same m=4 recipe does not memorize (index 0.26) and covers 0.97 of the modes (Table 16, Appendix F.3), and with m=16 the index stays at 0.29 even when training runs 1.5× longer. The budget, not the recipe, determines whether the model memorizes.

Excess memorization collapses onto $I(m, K)$. Proposition 1 says that the incentive is the codebook's miss distortion. We trained IMLE-WM for $K \in \{4, 8\}$ and $m \in \{4, 8, 16, 32\}$ (two seeds, 8k steps), fitted excess memorization as a linear function of $I(m, K)$ on those eight points ($0.015 + 1.06\, I$, RMSE 0.032), recorded predictions with ±2 RMSE intervals for nine runs at $K \in \{2, 16\}$, and only then trained those runs. Seven of the nine fell inside their interval (Figure 1a; Table 10, Appendix F). The two misses lie at the ends of the range and in the directions we had noted as risks: at K=2, m=2 ($I = 0.50$, outside the fitted range) memorization is stronger than the linear extrapolation (0.96 vs. 0.77), and at K=16, m=4 it is weaker (0.22 vs. 0.31), consistent with a capacity effect, since storing one of 16 outcomes per context costs $\log_2 16$ bits. The alternative hypothesis, that the budget should scale with the number of modes, is rejected:

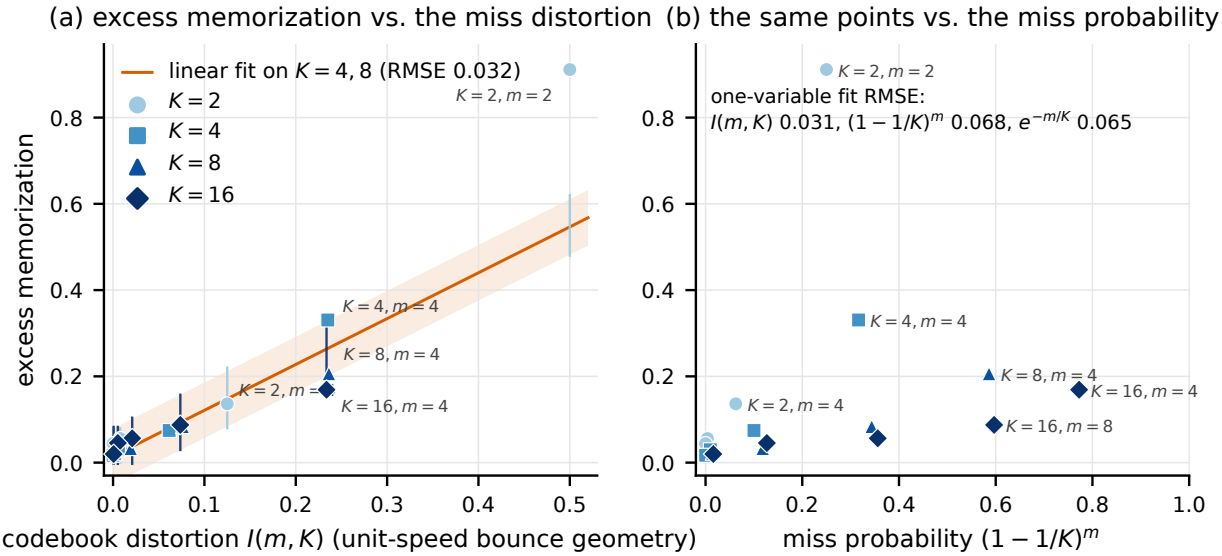


Figure 1: Excess memorization on A2-div at 8k steps (two seeds) for $K \in \{2, 4, 8, 16\}$, $m \in \{2, \ldots, 64\}$. (a) Against the codebook distortion $I(m, K)$ of Equation 3 (exact for the bounce geometry); line and band: the linear fit on the K=4, 8 points (± 2 RMSE) from which the K=2, 16 predictions were recorded before those runs; vertical bars: the recorded intervals. (b) The same points against the miss probability $(1 - 1/K)^m$ used by Gladstone et al. (2026) and Bennett (2026): no collapse.

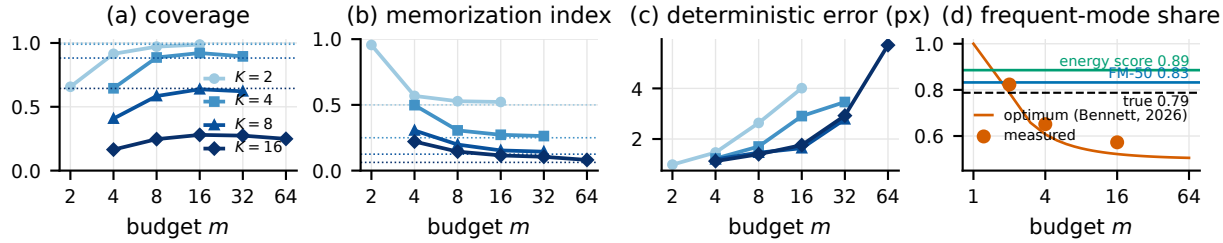


Figure 2: What the budget controls (A2-div, 8k steps, two seeds; shade = K). (a) Coverage peaks at m=16 (dotted: ideal sampler) while (b) the memorization index approaches chance (dotted). (c) The deterministic-transition error grows with m; training time grows about threefold from m=4 to m=32, inference cost not at all (all panels in Appendix F). (d) With unequal mode probabilities (K=2, p=0.79) the frequent-mode share of IMLE-WM samples falls with m along the optimum of Bennett (2026); FM-50 is calibrated and the energy score over-represents the frequent mode (Section 8).

at a fixed m=8 the excess memorization is 0.075, 0.083 and 0.087 for $K = 4$, 8 and 16, as the nearly equal values of $I(8, K)$ on this geometry predict, whereas at a fixed ratio $m/K = 1$ it ranges from 0.91 to 0.06; a one-variable linear fit on all points except the extrapolated one has an RMSE of 0.031 in $I(m, K)$, against 0.068 in $(1 - 1/K)^m$ and 0.065 in $e^{-m/K}$ (Figure 1b). The linear relation holds for $I \lesssim 0.25$ and saturates above it. The practical rule is that m must make $I(m)$ small relative to the mode spacing on the outcome manifold; here $m = 16$–32 brings the excess memorization below 0.05 for every K.

Countermeasures and the cost of m. Raising m is the cleanest way to remove the incentive (Table 11, Appendix F): RS-IMLE rejection of candidates already within ϵ of the target (Vashist et al., 2024) changes nothing at m=4, and Gaussian noise on the context tokens ($\sigma = 1.0$) removes the key and the memorization (index 0.27) but blurs information the prediction needs (coverage 0.74, precision 0.61, deterministic-transition

error 1.92 px). Figure 2 shows the costs of m itself: at 8k steps the deterministic-transition error grows from 1.2 px at m=4 to 2.8–3.5 px at m=32 (less with longer training), training time grows about threefold, and coverage and precision saturate at m=16. The recipe is otherwise insensitive: replacing Geman–McClure by ℓ_2, or the 256-dimensional noise by a 64-dimensional one, changes coverage and the index by at most 0.01 (Appendix D).

A strictly proper score memorizes regardless of m. Proposition 3 predicts that an energy-score generator memorizes in non-repeating contexts at every budget. Trained on A2-div with the IMLE-WM generator unchanged, it does (Table 2; Table 16 in Appendix F.3): the index reaches 0.87 at m=8 and 0.97 at m=16 (chance 0.25); its held-out coverage peaks at 0.50 near 6k steps and then falls to 0.36 as the index rises; and a larger m makes it memorize *faster* (index 0.84–0.87 at 8k steps for the two m=16 seeds, against 0.51–0.72 for the three m=8 seeds), as expected if m only reduces the gradient variance towards the same minimizer. A per-token energy score, a common choice for continuous tokens, memorizes as well (index 0.97, coverage 0.22). The ordering across objectives follows the relative incentives of Section 4 (regression 0.99, energy score 0.87–0.97, nearest-of-m 0.75 at m=4 and 0.29 at m=16), and with unequal mode probabilities the energy score has no calibration advantage either (frequent-mode share 0.89 against a true share of 0.79, coverage 0.58; Figure 2d).

Memorization requires an incentive, an identifying key and enough revisits of each context. A toy two-branch problem after Makabe et al. (2026) with a 16-dimensional nuisance "appearance" vector in the context reproduces their result when every training context is fresh (the energy score places 0.94 of its samples on a mode, nearest-of-m 0.92; they report 0.962 and 0.919), but with 1,024 fixed training contexts the energy score memorizes completely (index 0.999) and covers both modes on almost no new context, whereas nearest-of-m stays at 0.51 and covers 0.99–1.00; removing the appearance vector, or using 16,384 or more contexts, removes the energy score's memorization as well (Appendix F.3). Both conditions transfer to the full environment: the five predictions recorded before the A2-fix and per-token runs (the energy score does not memorize on A2-fix but still trails IMLE-WM in coverage and precision; the per-token variant memorizes on A2-div) all held, and on A2-fix the coverage gap shrinks from 0.57 to 0.04 while a precision gap of 0.14 remains. A second cause is spread off the manifold. On Bouncing Shapes, where nothing can be memorized, the energy score still trails IMLE-WM and regression (F1 8.9 vs. 7.2 and 8.1 px); its error rises to 33.8 px under FlowWM's clean heatmap readout and falls back to 11.7 px under a readout trained with feature noise 0.5, whereas IMLE-WM and regression move by less than 0.5 px; and on A2-div its token-space spread exceeds that of IMLE-WM while its readout spread is less than half (Appendix E). The diversity is spent in directions that readouts do not see, as the direction-insensitivity argument suggests. The conclusion is conditional: with fresh contexts or without an identifying feature the energy score is competitive, but in non-repeating contexts with identifying appearance, the property that Makabe et al. (2026) list as a defect of the hard minimum, namely that it is not strictly proper for finite m, is precisely what lets nearest-of-m escape memorization by raising m.

8 Results III: Where No Candidate Budget Suffices

Mode-probability flattening. On a bounce variant with K=2 modes of probability 0.8 and 0.2 (realized frequency 0.787 on the evaluation contexts), the share of the frequent mode among 64 samples per context falls with m as the stationarity condition of Bennett (2026) prescribes: 0.82, 0.65 and 0.57 for m = 2, 4 and 16, against theoretical values of 0.79, 0.61 and 0.52 (Figure 2d; Table 12, Appendix F). Every method, including FM-50 (0.83), measures about 0.04 above its reference value, a common offset of the readout-based assignment; relative to FM-50 the measured flattening is -0.18 and -0.26, against theoretical -0.18 and -0.27. The trade-off is real: m=2 is calibrated but memorizes (index 0.93) and covers only 0.75; m=16 covers 0.99 but reports the modes as nearly equiprobable; FM-50 does both; and the energy score does neither (share 0.89, coverage 0.58). Inverting the stationarity condition to recover p is ill-conditioned for large m (errors in the log-odds are amplified about 60× at m=16), so a large budget is a coverage device, not a probability estimate; for equiprobable modes, the calibration is exact (Section 9).

Under-dispersion on real robot video (BAIR). BAIR (Ebert et al., 2017) has no enumerable modes; its uncertainty is a continuous, high-dimensional variation of the arm and the objects. IMLE-WM is the most

Table 5: Per-step matching on self-rollouts (H=4) against one-step matching on stochastic Push-T at an equal training budget (8k steps; 21M, three seeds): precision / recall error (px) and the paired per-seed change.

Stochastic Push-T	h=8 prec. / recall	h=16 prec. / recall	Train (h)
One-step matching	25.7 / 25.4	41.6 / 43.7	1.3
Per-step, feed matched	**18.8** / **18.4**	**24.6** / **29.3**	5.3
Paired change	−27% / −28%	−41% / −33%	×3.9
One-step matching, 12k steps	21.8 / 22.1	34.1 / 37.8	2.2

accurate model there (best-of-16 token error 0.527 at h=16 against 0.843 for FM-50; manifold distance 1.30, like regression, against 1.78) but loses clearly on distribution metrics (Fréchet distance 2.09 vs. 1.38 and kNN recall 0.15 vs. 0.84 for FM-50; Table 13, Appendix F). The samples are under-dispersed: their spread-to-error ratio at h=1 is 0.22 at m=4 and 0.45 at m=16, the same on training and evaluation contexts, so this is not memorization. Raising the budget to m=64 moves the ratio only to 0.54 at 2.4× the training time, and matching in a 512-dimensional pooled space lowers it to 0.31–0.38; no one-step IMLE-WM variant closes the gap. Proposition 2 reproduces the growth of the spread with m using one fitted parameter ($d_{\text{eff}} \approx 27$, fitted at m=16, predicts 0.27 and 0.58 at $m = 4$ and 64 against measured values of 0.22 and 0.54; Figure 6a, Appendix F), but our recorded attempt to remove that parameter failed: the participation ratio of the regression residuals is about 1,030, far outside a factor of two of 27, and would predict a spread five times smaller than observed. An exploratory analysis designed after this failure suggests why: per context, the samples occupy only 1.2, 1.8 and 2.5 effective directions at $m = 4$, 16 and 64; these directions differ across contexts; and they align with the true residual of their own context ten times better than with that of another context, while capturing only 9–21% of it (Figure 6b,c). The contraction is therefore a collapse onto a few context-specific directions whose number grows slowly with m, the shape of reverse water-filling in rate–distortion theory; with a single true future per context we could not test this quantitatively.

9 Results IV: Rollouts and Planning

Rollouts stay on the token manifold. A one-step model is applied recursively, so beyond one step what matters is whether its outputs stay within the distribution of inputs it was trained on. On A2-div, the manifold distance of IMLE-WM with m=16 grows from 0.86 at h=1 to 1.68 at h=32, within 0.1 of regression (1.80), whereas FM-50 reaches 6.16, FM-2 6.36 (so the drift is not an integration error) and iMF 8.04 (Figure 7, Appendix F); the ordering holds on Bouncing Shapes (2.08 vs. 2.95–3.57 at h=16) and on Push-T (3.28 vs. 4.37–5.81). This explains why the autoregressive models in Table 3 are nearly indifferent to the readout while the joint-prediction models lose a factor of 2.7–4.4 under the heatmap readout, and why flow matching has the better best-of-16 error on Push-T alone: its rollouts spread more as they drift (36 vs. 22 px at h=16), so one of them lands closer (Appendix F). A readout trained with feature noise 0.5 rules out readout fragility as the cause: under it, regression and IMLE-WM improve (22.5 → 9.7 and 9.0 → 5.3 px, joint protocol) while flow matching worsens (12.6 → 15.0).

Per-step matching and what the fed-back state does. Per-step matching with H=4 reduces the h=16 recall error on stochastic Push-T from 43.7 ± 0.9 to 29.3 ± 2.8 px (hierarchical bootstrap: −14.3 $[-19.1, -9.9]$ px) and the precision error by 41%, and it beats the one-step-matched model trained 1.5× longer (Table 5). On the deterministic Push-T control, where the only train–test gap is exposure bias, the best-of-16 error at h=16 falls by 19%, so most of the gain comes from training on self-generated context; on A2-div only the manifold distance improves, and on A1 the best-of-16 rollout error improves by 24% with F1 unchanged (Appendix F.4). The cost is 3–4× the training time. Changing only the fed-back state separates the two mechanisms (Table 17 and Figure 9, Appendix F.4). Feeding back the true frame, with the same number of matched updates, changes nothing. Feeding back a random candidate keeps the rollout on the manifold at least as well as the matched candidate does (1.81 vs. 1.92) but does not improve the best-of-16 error (22.9 vs. 23.2) and halves the spread relative to the true futures: a candidate that has followed a different future makes the true next frame an unreachable target, so the one-step distribution widens into jitter that does not accumulate into distinct futures. Only the matched candidate keeps the context on the

future that was realized (1.11× the true spread; best-of-16 error 17.7 px). The manifold benefit thus comes from any self-generated context, and the coverage benefit from feeding back the candidate nearest the truth, which in nearest-of-m training is the argmin of the loss; the ablation was run on Bouncing Shapes only.

Planning: equal success, 50× faster, accurate risk estimates. On stochastic Push-T, IMLE-WM, regression and FM-2 are indistinguishable at 32–35%, FM-50 is at 30% on ten episodes, and the planning time per step is 4.0 s for IMLE-WM, half that of FM-2 and 1/50 that of FM-50 (Table 18, Appendix F.5); the shared contact error likely prevents every model from exploiting the hidden center of gravity. The branch-risk task isolates the question: the straight path to the goal crosses an ice patch along an absorbing pit, each push on the ice is deflected by $-60°$, $0°$ or $+60°$ with equal probability, and the upward deflection leads into the pit. IMLE-WM estimates the risk correctly (a one-push pit probability of 0.36 ± 0.03 against the true 0.33, and a five-push probability of 0.44 against 0.43), FM-2 under-estimates it by a factor of 2.4 because two Euler steps do not resolve the modes (with 50 steps it gives 0.31), and regression cannot represent it at all. Yet the IMLE-WM planner succeeds no more often than the regression planner (55% vs. 56%) and falls into the pit *more* often (31% vs. 24%), while the FM-2 planner, with the worst risk estimates, succeeds most often (67%; the oracle planner reaches 76%). Planning performance thus does not follow the model's probabilistic accuracy: a mean-distance cost over a few imagined futures lets the cross-entropy method exploit model errors, and an absorbing failure is not penalized much more than a detour, consistent with Cheung & He (2026), Terver et al. (2025) and Lee et al. (2026a). A planner that uses the estimated mode probabilities directly (through a conditional value-at-risk objective or a failure-probability constraint) is the natural next step; a mean-plus-variance cost is not, since it lowered the success rate from 78% to 50% even with the true simulator as the model (Appendix G).

10 Discussion and Limitations

When to use a one-step nearest-of-m world model. If the outcome uncertainty consists of a few discrete alternatives, the random codebook covers them: $I(m)$ vanishes exponentially once m exceeds the number of modes on the outcome manifold, and a budget of 16–32 removes the memorization incentive. If the uncertainty is continuous and high-dimensional, no affordable m covers it, and the samples collapse onto a few context-specific directions. A practical rule is to estimate $I(m)$ from the outcome diversity of near-identical contexts and to pick the smallest m at which it is a few percent of the mode spacing; a hybrid that selects the mode with nearest-of-m and models the residual with a few-step flow could serve both regimes.

Limitations. (1) The approach under-disperses on real video, and no one-step variant fixed this; the isotropic fit on BAIR uses one free parameter. (2) On stochastic Push-T the advantage over flow matching lies in precision; the recall advantage does not exceed seed variability with three seeds. (3) The benchmarks are synthetic or simulated except for BAIR, the models are small, and the flow baselines are far from the scale of Porcher et al. (2026). (4) Planning success did not improve on either task. (5) Large budgets flatten mode probabilities, and the energy score did not provide calibration here. (6) The ablations, the BAIR variants and the deterministic Push-T control use a single seed; the $K{=}2, 16$ grid and the energy-score follow-up runs use two. (7) Proposition 3 was derived after we observed the energy score memorizing; only the follow-up runs had predictions recorded in advance, and we examined a single β and two distance choices.

11 Conclusion

A one-step world model trained with a nearest-of-m objective on frozen DINO tokens covers enumerable futures better than multi-step flow matching and other one-step generators at matched parameter counts, with one network evaluation per step and rollouts that stay on the data manifold. The budget it needs is set by the distortion of a random codebook of m draws: a few dozen candidates suffice for mode-like uncertainty, and no budget does for continuous high-dimensional uncertainty.

Broader Impact Statement

This work studies the training objectives of small world models on synthetic, simulated and public robot data; we see no risks beyond those of world modeling in general, with one caution: samples from a nearest-

of-m model with a large budget over-represent rare modes (Section 8) and must not be read as calibrated probabilities in safety-relevant use.

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A Derivations

Proposition 1. With non-repeating contexts, the empirical loss of a conditional generator decomposes over training contexts. For the memorizing solution every candidate equals the realized mode, so the minimum distance is zero for every context. For the generalizing solution the candidates are m i.i.d. draws from $p(\cdot \mid c_i)$, and the realized mode is itself a draw from p, so the expected minimum distance is $I(m)$ of Equation 3. The difference of the two empirical risks is therefore $I(m)$. For the bounds, let the modes be equiprobable with pairwise distances in $[\delta, \Delta]$. The probability that none of the m draws equals the realized mode k is $(1 - 1/K)^m$; on that event the nearest draw is at squared distance in $[\delta^2, \Delta^2]$, and otherwise the distance is zero, which gives the stated bounds. If the modes densely fill a curve of length ℓ ($K \gg m$), the draws are

m uniform points on the curve and the expected squared arc distance from a uniform point to its nearest draw is $\Theta(\ell^2/m^2)$, as for the nearest of m uniform points on an interval, so $I(m)$ decays polynomially and does not depend on K. For squared-error regression the single prediction can memorize the realized mode at zero loss, and the generalizing prediction (the conditional mean) has loss $\operatorname{tr}\operatorname{Cov}_p$, independent of any budget. A detail used in Section 7: the memorization index s is related to the excess memorization by $s = 1/K + (1 - 1/K)\,\text{excess}$, so a ± 2 RMSE interval on the excess scale maps to $\pm 2(1 - 1/K)$ RMSE on the index scale.

Proposition 2. Write $Y_j - X = (\mu - X) + s\,\varepsilon_j$ with $\varepsilon_j \sim \mathcal{N}(0, I_d)$ and $X - \mu \sim \mathcal{N}(0, \sigma^2 I_d)$. Then $\|Y_j - X\|^2 = \|X - \mu\|^2 + s^2\|\varepsilon_j\|^2 - 2s\,\varepsilon_j^\top (X - \mu)$. The first term is common to all j; the second concentrates at $s^2 d$ with fluctuations of order $s^2\sqrt{d}$; the third is, conditionally on X, Gaussian with variance $4s^2\|X - \mu\|^2 \approx 4s^2\sigma^2 d$ and independent across j, so the minimum over j of the sum is, to leading order in d, $d\sigma^2 + ds^2 - 2s\sigma\sqrt{d}\,e_m$ with $e_m = \mathbb{E}\max_{j \le m} \xi_j$. Minimizing in s gives $s^\star = \sigma e_m/\sqrt{d}$. The spread-to-error ratio used in Section 8, defined as the mean pairwise distance among samples divided by the mean distance from samples to the truth, is $r = \sqrt{2}\rho/\sqrt{1 + \rho^2}$ with $\rho = s/\sigma$ under this model, which is how d_{eff} is fitted from one measured r.

Flattening. Let the generator place mass q_k on mode k and let all modes be at mutual distance Δ. The probability that none of m draws lands on the realized mode k is $(1 - q_k)^m$, so $L(q) = \Delta^2 \sum_k p_k (1 - q_k)^m$. Minimizing subject to $\sum_k q_k = 1$ gives the stationarity condition $m\,p_k(1 - q_k)^{m-1} = \lambda$ for every k with $q_k > 0$, which is Theorem 7 of Bennett (2026); for $K = 2$ it reads $\big((1 - q)/q\big)^{m-1} = (1 - p)/p$. If $p_k = 1/K$ for all k the symmetric point $q_k = 1/K$ satisfies the condition and is the unique minimizer of the convex objective for $m \ge 2$ (Theorem 4 of the same work). The numerical optima quoted in the text (for $p = (0.787, 0.213)$: 0.787, 0.607, 0.522 at $m = 2, 4, 16$) solve this equation.

Proposition 3. (a) The energy distance between q and δ_x is $2\mathbb{E}_{Y \sim q}\|Y - x\|^\beta - \mathbb{E}_{Y, Y' \sim q}\|Y - Y'\|^\beta - 0$, so $\mathrm{ES}_\beta(q, x)$ is half of it; for $\beta \in (0, 2)$ the energy distance is non-negative and zero only when the two distributions coincide (its square root is a metric on distributions with finite β-th moments; Gneiting & Raftery, 2007; Dawid, 2007), so it vanishes only at $q = \delta_x$. The empirical risk over training contexts is a sum of such terms with one x per context, so its unique minimizer is the memorizing solution; the m-sample estimator with the unbiased pairwise term has this risk as its expectation for every $m \ge 2$. (b) At $q = p$, $\mathbb{E}_{x \sim p}\,\mathrm{ES}_\beta(p, x) = \mathbb{E}\|Y - x\|^\beta - \frac{1}{2}\mathbb{E}\|Y - Y'\|^\beta = \frac{1}{2}\mathbb{E}\|Y - Y'\|^\beta$ because x and Y are exchangeable. (c) For $\beta = 2$, $\mathbb{E}\|Y - x\|^2 - \frac{1}{2}\mathbb{E}\|Y - Y'\|^2 = \|\mathbb{E}Y - x\|^2$. (d) The per-token score is a sum over tokens of energy scores of the token marginals, and (a)–(b) apply to each term.

Direction insensitivity (interpretation). Suppose the generator's mean misses the outcome by an unknown vector of length a and it adds isotropic noise of scale s in d dimensions with d large. Then $\mathbb{E}\|Y - x\|^\beta \approx (a^2 + s^2)^{\beta/2}$ and $\mathbb{E}\|Y - Y'\|^\beta \approx (2s^2)^{\beta/2}$, and the energy score $(a^2 + s^2)^{\beta/2} - \frac{1}{2}(2s^2)^{\beta/2}$ is minimized at $s = a$ for every $\beta \in (0, 2)$, where it equals $2^{\beta/2 - 1}a^\beta$; this is the $\lambda = 1$ case of Proposition 4.1 of De Bortoli et al. (2025), and the gain does not depend on d. The same calculation for the nearest-of-m loss follows Proposition 2 and gives $s \approx a\,e_m/\sqrt{d}$. The approximation requires the bias and the noise to be spread over many dimensions (participation ratio $\gg 1$); the regression residual on BAIR has participation ratio about 1,030, and we did not measure it on the bounce tokens. The argument says nothing about where the isotropic spread points; the claim in Section 7 that it points off the manifold is an empirical observation.

B $I(m, K)$ for the bounce geometry and the recorded predictions

At a wall frame the K possible next directions are the unit vectors at angles $\theta_k = \big((k + \frac{1}{2})/K\big)\,180^\circ - 90^\circ$ about the wall normal, $k = 0, \ldots, K - 1$, with unit speed. For a target mode k let $d_{(1)}^{(k)} \le \cdots \le d_{(K)}^{(k)}$ be the sorted squared distances from μ_k to all modes ($d_{(1)}^{(k)} = 0$). The nearest of m i.i.d. uniform draws is the r-th

closest mode with probability $\big((K-r+1)/K\big)^m - \big((K-r)/K\big)^m$, so

$$I(m,K) = \frac{1}{K}\sum_{k=1}^{K}\sum_{r=1}^{K} d^{(k)}_{(r)}\Big[\big(\tfrac{K-r+1}{K}\big)^m - \big(\tfrac{K-r}{K}\big)^m\Big]. \tag{4}$$

Predictions tested in the paper. P1: excess memorization collapses across K against $I(m,K)$, not against $(1-1/K)^m$ (K=2, 16 predictions recorded in advance; Section 7). P2: the energy score memorizes in non-repeating contexts at every m, faster at larger m, and stops when contexts carry no identifying feature (Section 7). P3: with $p = (0.8, 0.2)$ the frequent-mode share of IMLE-WM samples falls with m as prescribed (Section 8). P4: on real video the spread grows with m only like e_m and m=64 does not reach FM-50's distribution metrics (Section 8). P5: the energy score's spread is off-manifold, so its deficit survives a noise-robust readout (Section 7).

Recorded predictions for the K=2, 16 grid. Table 6 lists the values together with the predictions recorded before the $K \in \{2, 16\}$ runs. The predictions were made by fitting excess memorization linearly in I on the eight $K \in \{4, 8\}$ points (excess $= 0.015 + 1.06\, I$, RMSE 0.032) and mapping to the index scale; the intervals are ± 2 RMSE. We noted two risks at the time: the K=2, m=2 point lies outside the fitted range of I, and large K might memorize less than predicted because more bits must be stored per context. Both misses are of these kinds.

Table 6: Recorded predictions for the memorization index at 8k steps and the observed two-seed means. The two misses are marked.

K	m	$I(m,K)$	Chance	Predicted [interval]	Observed
2	2	0.500	0.50	0.77 [0.74, 0.81]	0.956 (miss)
2	4	0.125	0.50	0.57 [0.54, 0.61]	0.568
2	8	0.008	0.50	0.51 [0.50, 0.54]	0.528
2	16	≈ 0	0.50	0.51 [0.50, 0.54]	0.522
16	4	0.234	0.0625	0.31 [0.25, 0.37]	0.221 (miss)
16	8	0.074	0.0625	0.15 [0.09, 0.21]	0.144
16	16	0.021	0.0625	0.10 [0.06, 0.16]	0.115
16	32	0.005	0.0625	0.08 [0.06, 0.14]	0.105
16	64	0.0006	0.0625	0.08 [0.06, 0.14]	0.081
4	4 / 8 / 16 / 32	0.235 / 0.061 / 0.006 / 0.0001	0.25	(fitted)	0.498 / 0.306 / 0.272 / 0.263
8	4 / 8 / 16 / 32	0.236 / 0.076 / 0.019 / 0.002	0.125	(fitted)	0.305 / 0.198 / 0.154 / 0.144

C Benchmark construction

A2 appearance randomization. Each episode of A2-div draws a 22-dimensional appearance vector: background RGB in $[10, 110]$, ball RGB in $[150, 255]$, radius in $[3, 6]$ px, speed in $[10, 18]$ px per frame, and two rectangles with top-left corner coordinates in $[0, 48]$, side lengths in $[6, 20]$ and RGB in $[40, 200]$. Frames are rendered at 64×64 with an anti-aliased disc, upsampled to 224 px and encoded by DINOv2 ViT-S/14. Context novelty is the ratio between the median distance from a test context to its nearest training context and the median nearest-neighbor distance within the training set, both measured in a PCA-64 token space: 1.9 for A2-div and 0.35 for A2-fix. The evaluation uses 400 held-out wall contexts; a stochastic transition is one whose current frame touches a wall.

Push environment. The 64×64 arena contains an ice rectangle ($x \in [20, 44]$, $y \in [14, 40]$) and an absorbing pit ($x \in [24, 40]$, $y \in [0, 14]$). A push action is a 2-D vector; with probability 0.1 it is zero, and otherwise its direction is uniform and its magnitude is uniform in $[0.3, 1]$. The ball moves 10 px along the action direction rotated by the sampled deflection, and it respawns after four frames in the pit. The branch-risk task draws starting positions with $x \in [8, 12]$, $y \in [17, 21]$ and goals with $x \in [52, 56]$ in the same

y band. The planner uses CEM-MPC with horizon T=5, 64 candidate sequences, m_p=8 imagined futures per candidate (one for regression), and a cost equal to the mean distance between the decoded final position and the goal; 100 episodes per model and seed with a shared set of task seeds. The risk diagnostic evaluates 200 start states with the straight five-push action sequence against 256 true-simulator rollouts per start, and a one-push test from 200 ice states just below the pit.

Stochastic Push-T hidden variable. Table 7 shows the per-episode randomizations we tried before choosing the center of gravity; mass and friction have no effect under Push-T's quasi-static physics, and damping produces unrealistic sliding. The multi-future test set takes 128 starting states just before the first contact (an earlier version with starts further from contact did not diverge within 8 steps and was discarded), and for each start the 64 futures share the three context frames. Modes closer than 2 px in outcome are merged, leaving 3.9 distinguishable modes per start on average. The CEM planner for Push-T uses T=5, 64 candidates, m_p=4, 3 iterations with 8 elites, re-planning every step for at most 10 steps, with the mean token-space ℓ_2 distance to the goal as the cost, following Terver et al. (2025); goals are within 25 steps of the start, and success requires the block within 20 px and 20° of the goal.

Table 7: Candidate hidden variables for stochastic Push-T: deviation of the T vertices after 16 steps under the same actions (median / maximum over episodes; the nominal displacement is 124 px).

Randomization	Deviation (px)	Decision
Block mass 0.5–2×	0.0 / 0.0	no effect (quasi-static)
Contact friction 0.3–1.2	0.0 / 0.0	no effect
Damping 0–0.1	25 / 346	unrealistic sliding
Center of gravity, radius 20 px, 4 positions + 3 px jitter	28 / 138	adopted
Center of gravity, radius 25 px	56 / 195	too large

BAIR. We use the first part of the official archive: 4,096 training sequences and 1,024 held-out sequences (the official test set plus three held-out training files), 64×64 frames, 4-dimensional actions, 3 context frames and 16-step rollouts.

D Implementation details and ablations

Code, environments, the queue files that define every run and the per-run outputs behind every figure and table will be released with the paper.

Training. AdamW with learning rate 2×10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.9)$, no weight decay, a linear warm-up over 1,000 steps followed by a cosine decay, batch 128, gradient clipping at norm 1, an exponential moving average of the weights with decay $\min(0.999, (1+t)/(10+t))$ used for all evaluations, 12k steps (8k for the budget grid, the asymmetric variant and the per-step-matching comparisons; 36k for iMF; 24k for Shortcut; 48k for the four-times-longer A1 runs). The noise vector has 256 dimensions and is normalized to unit RMS before the mapping network. Late fusion sets the first five of eight blocks (eight of twelve for the 21M model) to a learned constant style. Candidate generation for matching runs without gradient tracking, in bfloat16, in a separate autocast region; if it shares a region with the gradient pass, autocast reuses the weight casts made without gradient tracking and silently cuts the gradient to the weights. The Geman–McClure scale is twice the running median (EMA 0.99) of the matching distance. The energy-score runs use $\beta = 1$, the whole-map RMS distance, the unbiased pairwise estimator, and backpropagate in two chunks of contexts; the per-token variant computes pairwise differences explicitly to be exact near zero. Flow matching uses the time shift $\alpha = 13$ at training and sampling and the Euler sampler. The iMF port follows the official JAX implementation through a PyTorch re-implementation: dual u/v heads of width 640 (to match parameters), the JVP direction taken from the model's own v prediction, and the loss $\|u + (t-r)\,\mathrm{sg}(\partial_t u) - v\|^2$ plus a flow-matching loss on v; it converges only with the time-embedding frequency scale set to 1 rather than the usual 1000. Our own earlier MeanFlow implementation diverged in three attempts once the average-velocity term

was switched on, and was abandoned. Shortcut models devote a quarter of each batch to the self-consistency target computed with the EMA model. Parameter counts: 8M setting, 7.97–8.24M across methods (8.24M for IMLE-WM); 21M setting, 20.6M for IMLE-WM. Training times in the tables are wall-clock hours on one RTX A5000; some runs shared a GPU and are therefore slower than the stand-alone timings (one training step at batch 128: 178 ms for m=4 and 488 ms for m=16 with the 8M model; 404 and 1112 ms with the 21M model).

Recipe choices. Table 8 gives the pilot comparisons behind the recipe. The wide head does not change IMLE-WM's results (it is there to match the flow baselines); late fusion halves training time at equal coverage and lowers the deterministic-transition error; the FlowWM recipe raised the flow baseline substantially. Multi-stage supervision, which ROMS-IMLE found important for images, does not help on a fixed 16×16 token grid (Appendix G).

Table 8: Recipe comparisons on A2-div (K=4, pilot and preparation runs, one seed; bootstrap 95% intervals over 400 contexts where given). Training hours are on a shared GPU.

Variant	Coverage	Precision	kNN recall	Det. err. (px)	Memo. (train / val)	Train (h)
IMLE-WM m=16, full noise injection, wide head	0.934 [0.917, 0.949]	0.83	0.94	2.31	0.29 / 0.24	1.93
IMLE-WM m=16, late fusion, wide head (adopted)	0.925 [0.907, 0.942]	0.83	0.95	1.56	0.29 / 0.27	1.05
IMLE-WM m=16, full injection, narrow head (6.6M)	0.930 [0.909, 0.948]	0.84	0.94	2.02	0.30 / 0.25	1.71
IMLE-WM m=16, pilot recipe (three spatial stages)	0.91 [0.89, 0.93]	0.81	0.94	2.67	0.30 / 0.22	3.2
FM, default recipe, 20 Euler steps	0.58 [0.55, 0.61]	0.38	0.68	4.64	–	0.6
FM, FlowWM recipe, 50 steps (adopted)	0.79 [0.77, 0.82]	0.51	0.91	2.27	0.29 / 0.24	0.4
iMF, time-embedding scale 1000			does not converge			
iMF, time-embedding scale 1, 12k steps	0.55 [0.52, 0.58]	0.22	–	13.1	–	1.0

Ablations. On A2-div with m=16 (one seed): replacing Geman–McClure by ℓ_2 gives coverage 0.93, precision 0.83, index 0.28, deterministic-transition error 1.81 px (vs. 1.55); a 64-dimensional noise vector gives 0.92, 0.83, 0.28, 1.50; training 1.5× longer (18k steps) gives 0.92, 0.86, 0.31, 1.45. On stochastic Push-T an 8M IMLE-WM with m=16 reaches recall errors 24.5 / 39.0 px at $h = 8$ / 16 against 22.1 / 37.8 for the 21M model (three seeds), so model size matters little there. In the pilot with the earlier recipe, Geman–McClure improved precision (0.92 vs. 0.86) and token-space coverage (0.85 vs. 0.66) over ℓ_2. The iMF learning curve on A2-div (one seed, checkpoints of a 72k-step run) saturates after 24k steps: coverage 0.56, 0.57, 0.58, 0.58 and precision 0.24, 0.25, 0.26, 0.26 at 24k, 42k, 60k and 72k steps.

E Readouts

Position readouts are small networks trained on real tokens: a soft-argmax probe for the ball (test error 0.13 px on A2-div, 0.05 px with DINOv3 on A1), and for Push-T a probe for agent position, block position and angle (T-vertex error 2.0 px). Because a probe might decode generated (out-of-distribution) tokens differently from real ones, the pilot cross-checked it against a nearest-neighbor judge that decodes a token map by looking up the nearest of 12,000 real frames in a PCA-128 space: on the fixed-appearance environment the two judges agreed within 0.01 for every IMLE-WM run (e.g. coverage 0.960 vs. 0.960 at K=4) and the nearest-neighbor judge was 0.03–0.09 lower for flow matching (0.841 vs. 0.814 with 20 steps), so the probe is, if anything, generous to the baselines; the ranking was identical under both. The FlowWM heatmap readout, ported from the public code, maps the 384-channel token map through a 1×1 convolution to 256 channels, a GELU, dropout 0.1 and a 16× transposed convolution to 256×256 logits per shape; it is trained with a pixel-wise cross-entropy and feature noise of standard deviation 0.05 for 20k steps and reads out the argmax; its error on real features is 0.10 px (95th percentile 0.75 px). The noise-robust variant is trained with feature noise 0.5 (error floor 1.03 px). Table 9 gives all readout combinations for A1, including the energy-score generator, whose rollout manifold distance at h=16 is 3.05 against 2.95–3.03 for flow matching and 1.98–2.08 for IMLE-WM.

Table 9: Bouncing Shapes F1 error (px) under three readouts, three seeds (energy score two). The last row is the 48k-step run of Section 6 (one seed).

	Joint 16→16			Autoregressive		
Method	probe	heatmap	heatmap, noise 0.5	probe	heatmap	heatmap, noise 0.5
Regression	5.14	22.48	9.67	8.09	8.41	8.53
FM-50	3.86	12.59	15.00	12.28	14.11	18.38
FM-2	3.83	14.80	16.43	9.95	12.49	16.85
Energy score, m=8	–	–	–	8.94	33.77	11.65
IMLE-WM, m=4	3.30	11.73	5.92	6.61	6.83	7.04
IMLE-WM, m=16	3.32	9.04	5.28	7.21	7.64	7.92
IMLE-WM, m=4, 48k steps	–	–	–	4.08	4.20	–

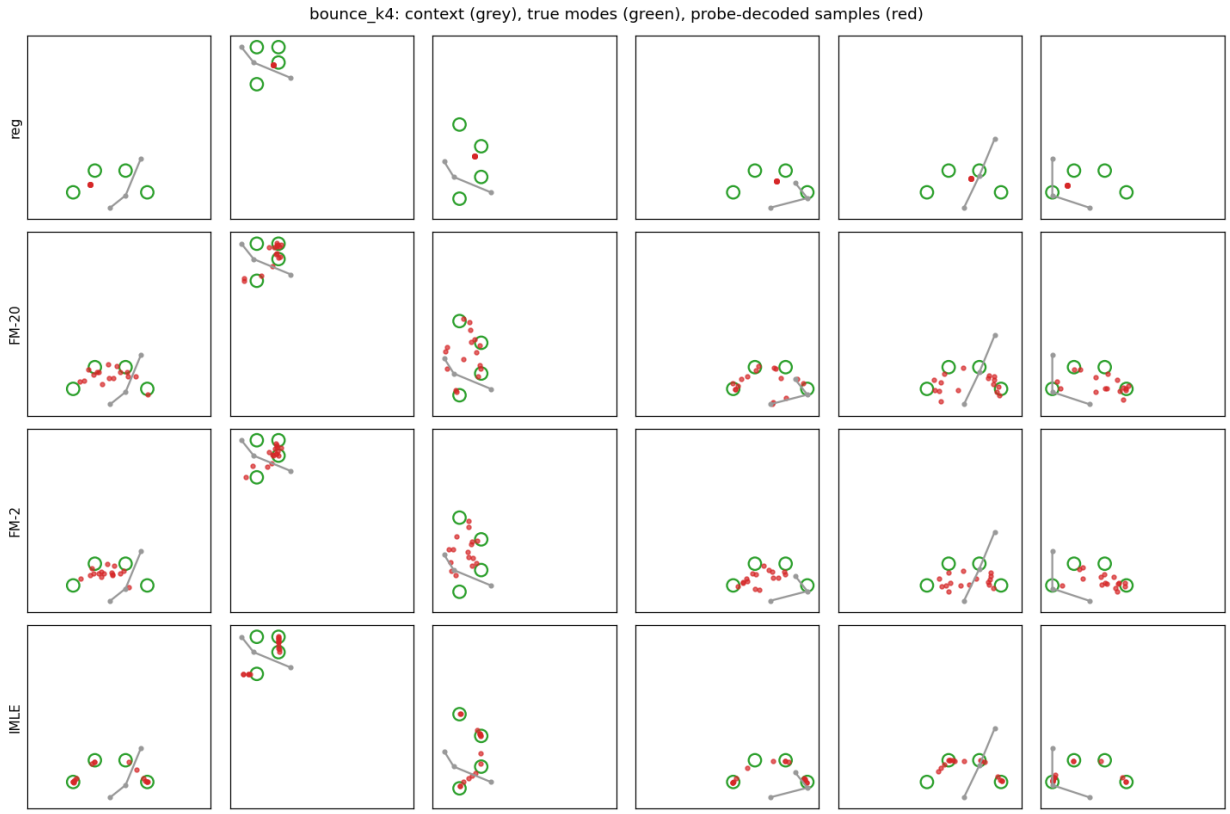


Figure 3: Qualitative pilot on the bounce environment with K=4 modes and fixed appearance (pilot recipe, m=4). Each column is one context whose ball has just hit a wall: gray, the three context frames; green circles, the four possible next positions; red, 16 samples decoded by the position readout. Regression (top) returns one point near the mean, flow matching with 20 or 2 Euler steps scatters between the modes, and IMLE-WM (pilot recipe) places its samples on the modes.

F Additional results

Memorization with longer training and larger models (pilot recipe, A2-div, K=4, one seed). With m=4, doubling the training steps (12k → 24k) raised the memorization index from 0.66 to 0.92 and lowered held-out coverage from 0.52 to 0.38; a 2.7× larger model (19M) at 12k steps reached 0.86 with

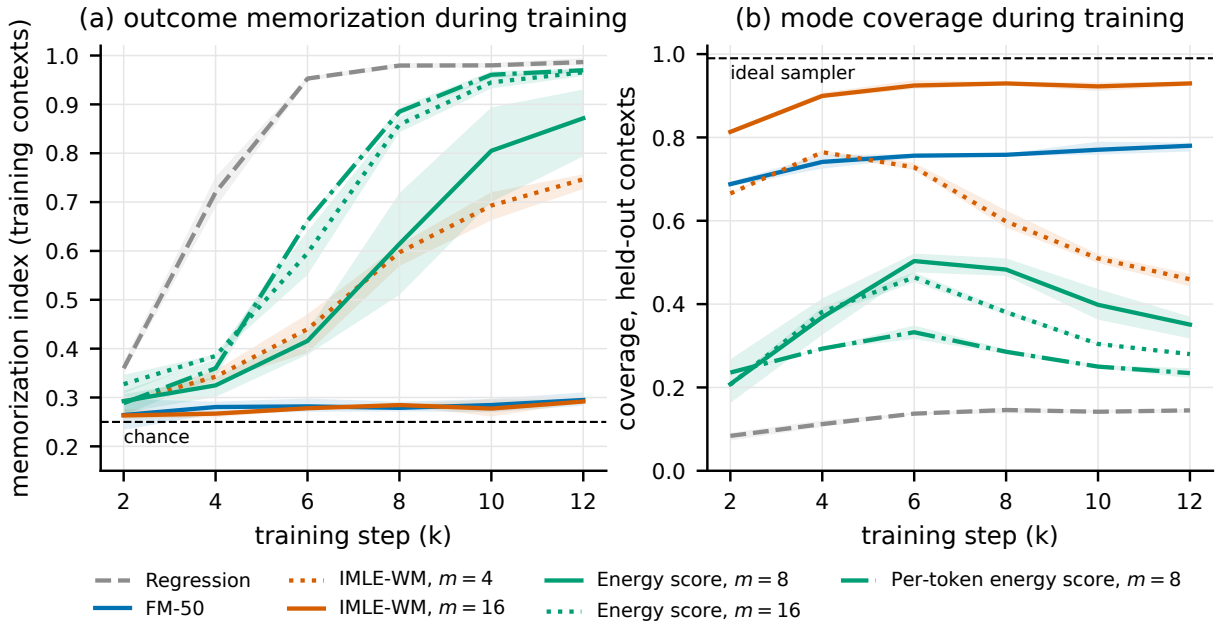


Figure 4: Memorization during training on A2-div (K=4; mean over seeds, band = min–max). (a) Memorization index (share of samples on the realized mode of training contexts); chance is 0.25 and held-out contexts stay at chance for every method. (b) Coverage on held-out contexts. Regression memorizes completely; IMLE-WM with m=4 and both energy-score variants memorize progressively and lose coverage as they do; IMLE-WM with m=16 and FM-50 do not memorize at this budget.

Table 10: Budget grid on A2-div at 8k steps, two seeds: memorization index (chance $1/K$), coverage, precision and deterministic-transition error (px). Coverage is capped by the ideal 16-sample sampler (1.00, 0.99, 0.88, 0.64 for $K = 2, 4, 8, 16$). K=16 is under-trained at 8k steps: its kNN recall rises from 0.12 to 0.85 with m while its coverage stays near 0.28.

	K=2				K=4				K=8			
m	2	4	8	16	4	8	16	32	4	8	16	32
Memo. index	0.96	0.57	0.53	0.52	0.50	0.31	0.27	0.26	0.31	0.20	0.15	0.14
Coverage	0.66	0.92	0.97	0.99	0.65	0.89	0.92	0.90	0.41	0.58	0.64	0.62
Precision	0.78	0.85	0.85	0.68	0.69	0.76	0.77	0.72	0.54	0.62	0.64	0.60
Det. err. (px)	0.99	1.47	2.64	4.01	1.23	1.70	2.91	3.46	1.20	1.46	1.64	2.78

	K=16				
m	4	8	16	32	64
Memo. index	0.22	0.14	0.12	0.11	0.08
Coverage	0.17	0.25	0.28	0.27	0.25
Precision	0.32	0.38	0.41	0.38	0.33
Det. err. (px)	1.13	1.40	1.76	2.92	5.71

coverage 0.43. On held-out contexts the index stayed at chance (0.23–0.25) in all three runs. With m=16, training 1.5× longer left the index at 0.31 (Appendix D).

Baseline fairness checks. iMF: a 72k-step run on A2-div (one seed, checkpoints every 6k steps) saturates after 24k steps at coverage 0.56–0.58 and precision 0.24–0.26, so its 36k-step budget in the main tables is not

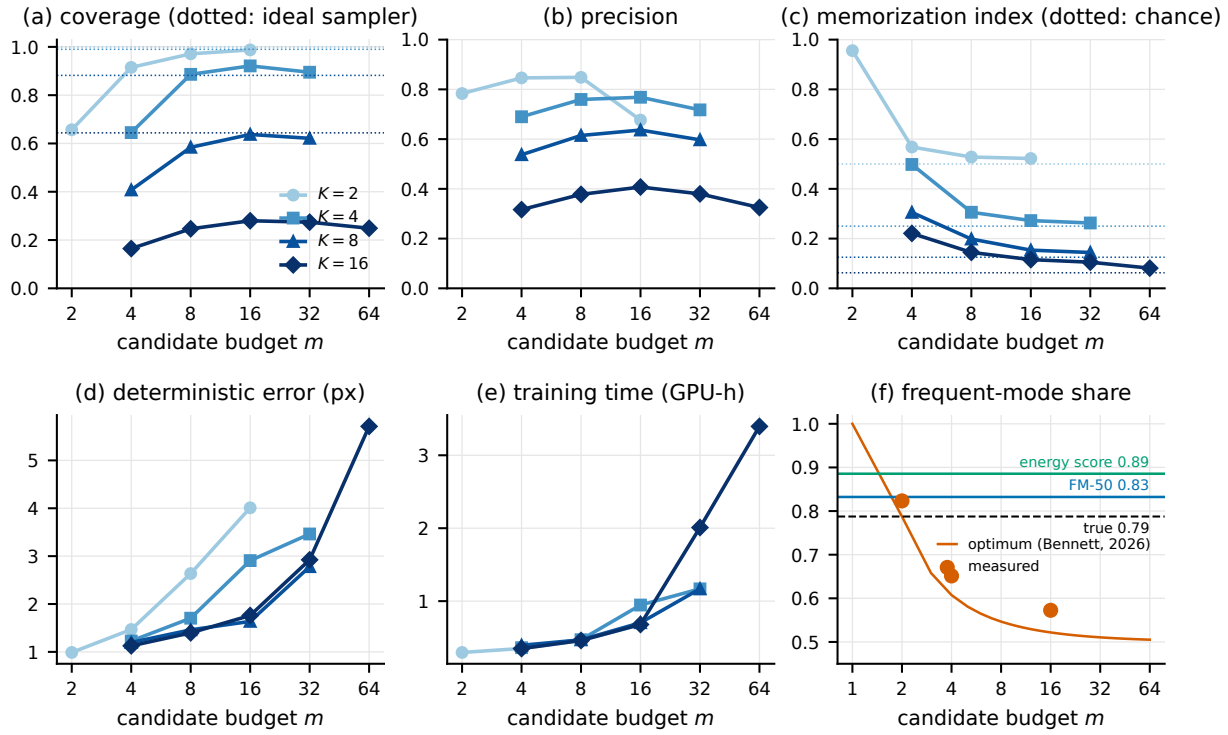


Figure 5: Full version of Figure 2: (a) coverage, (b) precision, (c) memorization index, (d) deterministic-transition error, (e) training time (GPU-hours for 8k steps) and (f) frequent-mode share on the asymmetric variant, all against the budget m (A2-div, 8k steps, two seeds).

Table 11: Countermeasures against outcome memorization on A2-div (K=4, 12k steps, two seeds; the m=4 and m=16 rows are the three-seed values of Table 2). Memorization index for training / held-out contexts.

Variant	Coverage	Precision	kNN recall	Memo. (train / held-out)	Det. err. (px)	Train (h)
m=4	0.46	0.68	0.65	0.75 / 0.25	1.14	0.5
m=4 + RS-IMLE rejection (ϵ=0.5)	0.46	0.69	0.65	0.74 / 0.25	1.17	0.5
m=4 + context noise σ=1.0	0.74	0.61	0.86	0.27 / 0.24	1.92	0.5
m=8 + context noise σ=0.3	0.88	0.77	0.92	0.32 / 0.22	1.65	0.7
m=16	**0.93**	**0.82**	**0.94**	0.29 / 0.25	1.55	1.1

the limiting factor. Flow matching: with four times the steps on A1 (48k, one seed) the F1 error becomes 4.08 (IMLE-WM m=4), 7.10 (FM-2) and 8.99 (FM-50) under the probe, and 4.20 for IMLE-WM under the heatmap readout; the FlowWM recipe (wide head, time shift 13, 50 steps) raised the A2-div coverage of our initial 20-step flow baseline from 0.58 to 0.79 and its precision from 0.38 to 0.51. Shortcut: trained for 24k steps so that its wall-clock time (0.7 h on A2, 1.5 h on B) is below that of IMLE-WM with m=16; sampling with two steps instead of one changed its metrics by less than 0.01. Parameter counts: 7.97–8.24M (8M setting), 20.6M (21M setting).

F.1 Rollout metrics

Table 14 completes Figure 7 with the best-of-16 readout error and the spread at the last horizon.

Table 12: Asymmetric bounce (K=2, $p = 0.787$ on the evaluation contexts, 8k steps, two seeds): share of the frequent mode among 64 samples per held-out context, its theoretical optimum, coverage of both modes, and the memorization index on training contexts (chance 0.5).

Method	Frequent-mode share	Theory	Share − FM-50 (measured / theory)	Coverage	Memo. index
IMLE-WM, m=2	0.823	0.787	−0.01 / 0.00	0.75	0.93
IMLE-WM, m=4	0.651	0.607	−0.18 / −0.18	0.96	0.68
IMLE-WM, m=16	0.573	0.522	−0.26 / −0.27	0.99	0.56
FM-50	0.832	0.787	–	0.95	0.73
FM-2	0.915	–	+0.08 / –	0.80	0.78
Energy score, m=8	0.886	0.787	+0.05 / 0.00	0.58	0.91

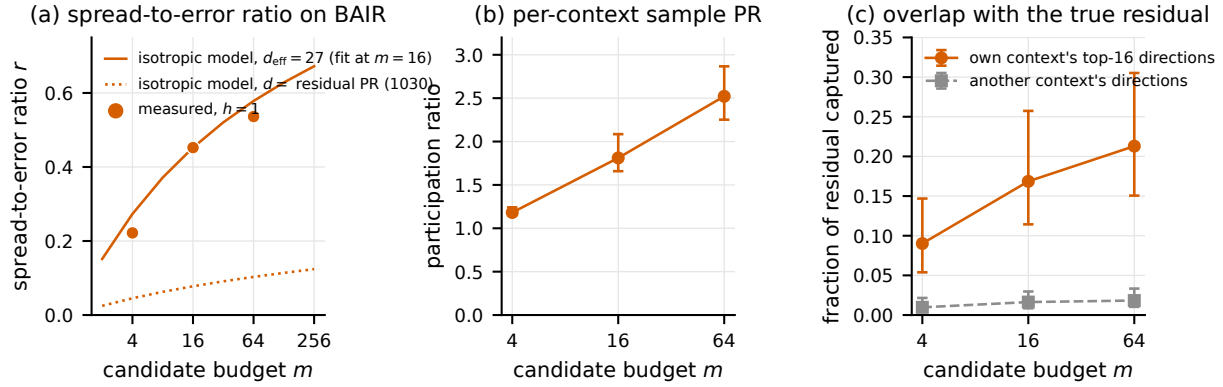


Figure 6: Under-dispersion on BAIR (h=1, 256 evaluation contexts). (a) Spread-to-error ratio against m: the isotropic model of Proposition 2 with one parameter fitted at m=16 ($d_{\text{eff}} \approx 27$) reproduces the m=4 and 64 points within 0.05; with d set to the measured participation ratio of the regression residuals ($\approx$ 1,030) it predicts a spread five times too small. (b) Participation ratio of the 64 samples of a context (median, quartiles): one to three directions. (c) Fraction of the true residual in a context's own top-16 sample directions versus another context's: the directions are context-specific and aligned with the truth, but capture 9–21% of it.

F.2 Stochastic Push-T: bootstrap intervals

Table 15 gives the paired differences behind Section 6 and Section 9.

F.3 Energy-score follow-ups

Figure 8 shows the toy experiment of Section 7. The task is the two-branch problem of Makabe et al. (2026) ($u \sim U[0.05, 1]$, $y = \pm\sqrt{u} + 0.08\varepsilon$, equiprobable modes) with a 16-dimensional nuisance vector $a \sim \mathcal{N}(0, I)$ appended to the context, a three-layer MLP generator of width 256 with a 4-dimensional noise input, Adam at 2×10^{-4}, batch 256, 15k steps, and the same energy-score estimator as the main runs. With fresh contexts at every step the energy score matches 0.940 / 0.946 of its samples to a mode center within 0.24 (m = 8 / 16) and nearest-of-m 0.916 / 0.924. With 1,024 fixed training contexts (about 3,750 revisits each) the energy score memorizes (index 0.999) and covers both modes on 0.2% of new contexts, while nearest-of-m with m = 8 / 16 is at 0.52 / 0.51 and covers 0.99 / 1.00. With 4,096, 16,384 and 65,536 contexts the energy score's memorization index is 0.73, 0.51 and 0.50; without the nuisance vector (context = u only) it is 0.50 even with 1,024 contexts. The thresholds are those of this small network and do not transfer; the 8M model on A2-div saw each of its 11,559 stochastic training contexts about 38 times and memorized.

Table 13: BAIR robot pushing, 21M models, two seeds (the m=64 and coarse-matching rows are one seed). Token RMS error of the best of 16 samples and of their mean, manifold distance and spread at h=16, Fréchet distance and kNN recall of mean-pooled PCA-32 tokens against 1,024 real frames.

Method	NFE	best-of-16 ↓ (h=1/8/16)			mean h=16 ↓	manifold h=16 ↓	spread h=16	FD h=8/16 ↓	recall h=8/16 ↑
Regression	1	0.268	0.481	0.556	0.556	**1.27**	0	1.44 / 2.65	0.21 / 0.08
FM-50	50	0.338	0.681	0.843	0.885	1.78	0.94	**0.53 / 1.38**	**0.73 / 0.84**
FM-2	2	0.302	0.590	0.734	0.756	1.45	0.72	1.42 / 2.75	0.26 / 0.08
Shortcut	1	0.401	0.902	1.173	1.197	1.71	1.48	0.95 / 1.77	0.42 / 0.26
IMLE-WM, m=4	1	0.260	0.460	0.532	**0.559**	1.28	0.18	1.35 / 2.35	0.24 / 0.12
IMLE-WM, m=16	1	**0.258**	**0.456**	**0.527**	0.576	1.30	0.29	1.22 / 2.09	0.28 / 0.15
IMLE-WM, m=64	1	0.257	0.461	0.534	0.589	1.33	0.32	1.23 / 2.00	0.28 / 0.14
IMLE-WM, m=16, coarse matching	1	0.259	0.463	0.535	0.565	1.29	0.21	1.21 / 2.03	0.33 / 0.15

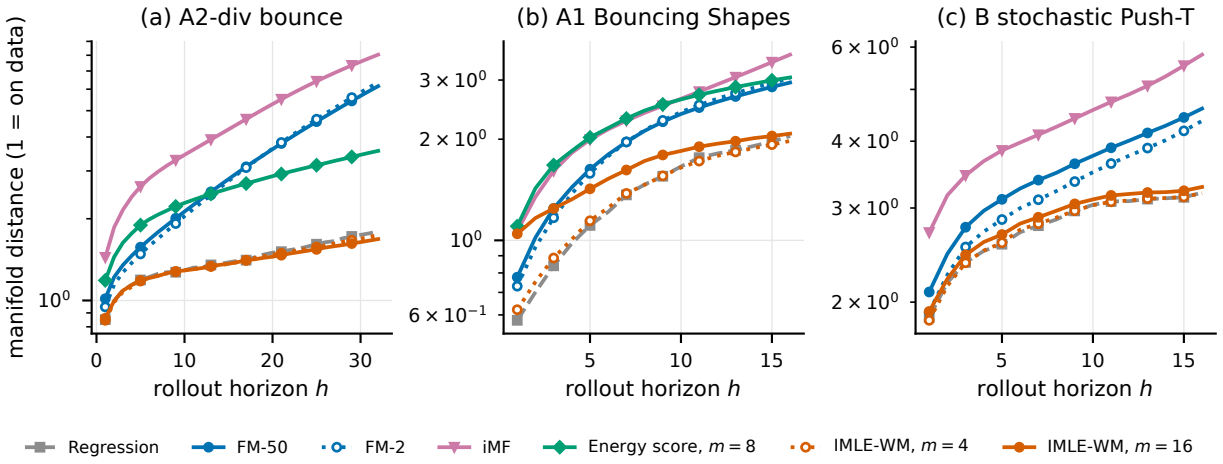


Figure 7: Manifold distance of autoregressive rollouts against the horizon (16 rollouts from real contexts, three seeds, log scale; 1 = on the data). IMLE-WM rollouts stay as close to the data as regression on all three benchmarks; flow-matching rollouts drift equally with 50 and with 2 integration steps; iMF and the energy score drift further.

The follow-up runs on A2-fix and with the per-token score were preceded by five recorded predictions, all of which held (two seeds each): (P1) the energy score with m=8 does not memorize on A2-fix (index ≤ 0.40; observed 0.268); (P2) its coverage on A2-fix exceeds its A2-div value but stays below IMLE-WM with m=16 in both coverage and precision (observed 0.947 vs. 0.987 and 0.814 vs. 0.954); (P3) IMLE-WM with m=4 does not memorize on A2-fix (≤ 0.35; observed 0.256); (P4) the per-token energy score memorizes on A2-div (index ≥ 0.7, coverage ≤ 0.5; observed 0.970 and 0.224); (P5) its one-step manifold distance exceeds 1.1 (observed 1.12; IMLE-WM 0.86). On A2-fix the energy score's kNN recall is 1.00 against 0.85 for IMLE-WM, so its samples spread more widely than the outcomes; its deterministic-transition error is the lowest (0.20 px). On A2-div its learning curves peak in coverage at 6k steps (0.50) and decline as the index rises ($0.42 \rightarrow 0.87$); the two m=16 seeds reach an index of 0.84 and 0.87 at 8k steps against 0.51–0.72 for the three m=8 seeds. Token-space diagnostics at h=1 on A2-div (three seeds, evaluation contexts): spread-to-error ratio 0.80 (energy score) vs. 0.64 (IMLE-WM 16), 0.38 (IMLE-WM 4), 1.15 (FM-50); best-of-16 token error 0.468 vs. 0.271; mean-prediction error 0.451 vs. 0.396; readout spread 1.7 vs. 3.9 px; one-step manifold distance 1.19 vs. 0.86.

Table 14: Rollouts from real contexts (16 samples, three seeds; iMF on Push-T two): manifold distance, best-of-16 readout error (px) and spread (px) at the last horizon (h=32 on A2-div, 16 on A1 and B).

	A2-div, h=32			A1 Bouncing Shapes, h=16			B Push-T, h=16		
Method	manifold	best-of-16	spread	manifold	best-of-16	spread	manifold	best-of-16	spread
Regression	1.80	32.2	0	2.04	34.0	0	3.21	102.2	0
FM-50	6.16	9.8	25.7	2.95	28.0	25.1	4.60	**67.4**	36.4
FM-2	6.36	10.6	26.8	3.03	23.4	26.1	4.37	69.2	29.7
iMF	8.04	10.1	29.0	3.57	33.4	–	5.81	84.1	57.0
IMLE-WM, m=4	1.74	10.5	23.9	**1.98**	**19.0**	–	**3.20**	88.7	9.2
IMLE-WM, m=16	**1.68**	**9.3**	27.2	2.08	19.4	17.9	3.28	79.1	21.8

Table 15: Paired differences on the stochastic Push-T multi-future test set (px; negative favors the first model) with 95% bootstrap intervals over the 128 starting states after averaging seeds, and hierarchical intervals that also resample the three seeds (10,000 resamples).

Comparison	Metric	Difference	Start-state CI	Hierarchical CI
IMLE-WM 16 vs. FM-50	h=8 recall	-1.8	$[-2.5, -1.1]$	$[-5.2, +2.3]$
	h=8 precision	-5.2	$[-5.7, -4.8]$	$[-8.0, -2.5]$
	h=16 recall	-4.2	$[-5.3, -3.1]$	$[-8.8, +1.1]$
	h=16 precision	-6.6	$[-7.6, -5.7]$	$[-8.5, -4.7]$
IMLE-WM 16 vs. FM-2	h=16 recall	-3.7	$[-4.9, -2.5]$	$[-8.7, +1.9]$
	h=16 precision	-5.6	$[-6.6, -4.6]$	$[-7.7, -3.2]$
IMLE-WM 16 vs. Shortcut	h=16 recall	-8.2	$[-10.0, -6.3]$	$[-11.3, -4.0]$
	h=16 precision	-13.2	$[-15.1, -11.3]$	$[-16.3, -10.2]$
Per-step vs. one-step matching (8k)	h=8 recall	-7.1	$[-8.2, -6.0]$	$[-9.2, -4.8]$
	h=16 recall	-14.3	$[-16.3, -12.5]$	$[-19.1, -9.9]$
	h=16 precision	-17.0	$[-18.4, -15.6]$	$[-20.4, -14.5]$

F.4 Per-step matching

Per-seed Push-T recall errors at h=16 (one-step → per-step): 43.7 → 30.3, 42.6 → 32.2, 44.7 → 25.5. On the deterministic Push-T control (one seed) the best-of-16 T-vertex error at $h = 4/8/16$ moves from 35.8/60.4/72.3 to 29.2/45.6/58.9 px and the manifold distance at h=16 from 3.90 to 3.64. On A2-div (two seeds, 8k steps) the manifold distance at $h = 8/16/32$ moves from 1.31/1.47/1.83 to 1.19/1.26/1.40 while one-step coverage and precision move from 0.92/0.77 to 0.90/0.73 and the deterministic-transition error from 2.91 to 3.97 px; the best-of-16 error is unchanged. On A1 (two seeds) F1 moves from 8.74 to 8.51 (precision 9.26 → 9.77, recall 8.28 → 7.55), the manifold distance at h=16 from 2.17 to 1.92 and the best-of-16 error from 23.2 to 17.7 px. The fed-back-state ablation was preceded by three recorded predictions: (S1) feeding back the true frame yields less than half of the manifold gain (manifold at h=16 $\geq$ 2.05; observed 2.14); (S2) feeding back a random candidate yields more than half ($\leq$ 2.05; observed 1.81); (S3) the random candidate's best-of-16 error exceeds the matched one's by at least 1 px ($\geq$ 18.7; observed 22.9). The spread of the 16 rollouts at $h = 1/4/8/16$ is 0.20/1.90/6.81/23.6 px for the true futures, 0.70/3.35/7.42/17.0 for one-step matching, 1.16/5.42/12.2/26.3 for the matched variant, 2.41/4.19/6.38/12.4 for the random variant and 0.61/3.07/7.08/16.6 for the true-frame variant. Training times: one-step 0.68 h, per-step variants 2.7 h (8M, 8k steps).

F.5 Planning details

On stochastic Push-T the final block position and angle errors are 46.6 px / 42.0° (regression), 40.6 / 35.2 (FM-2), 45.2 / 20.0 (FM-50, 10 episodes) and 49.5 / 44.9 (IMLE-WM). The one-step contact error (block displacement after a contact transition, true 15.9 px) is 11.7 / 11.2 px for regression at 12k / 36k steps, 12.4

Table 16: Energy score against nearest-of-m on bounce (12k steps; two seeds unless noted). A2-div: randomized appearance, contexts never repeat. A2-fix: fixed appearance, no identifying key (same recipe for all rows). The one-step manifold distance is in units of the median real-to-real distance (A2-fix: relative to IMLE-WM m=16). A1 and the readout comparison are in Appendix E.

Setting	Method	Coverage	Precision	kNN recall	Memo. index	Det. err. (px)	Manifold h=1
A2-div	Energy score, m=8 (3 seeds)	0.36	0.61	0.77	0.87	1.06	1.19
	Energy score, m=16	0.28	0.61	0.60	0.97	1.06	–
	Per-token energy score, m=8	0.22	0.55	0.44	0.97	1.33	1.12
	IMLE-WM, m=4 (3 seeds)	0.46	0.68	0.65	0.75	1.14	0.84
	IMLE-WM, m=16 (3 seeds)	**0.93**	**0.82**	**0.94**	0.29	1.55	0.86
A2-fix	Energy score, m=8	0.95	0.81	**1.00**	0.27	0.20	1.59×
	IMLE-WM, m=4	0.97	0.91	0.68	0.26	0.15	0.94×
	IMLE-WM, m=16	**0.99**	**0.95**	0.85	0.25	0.91	1×

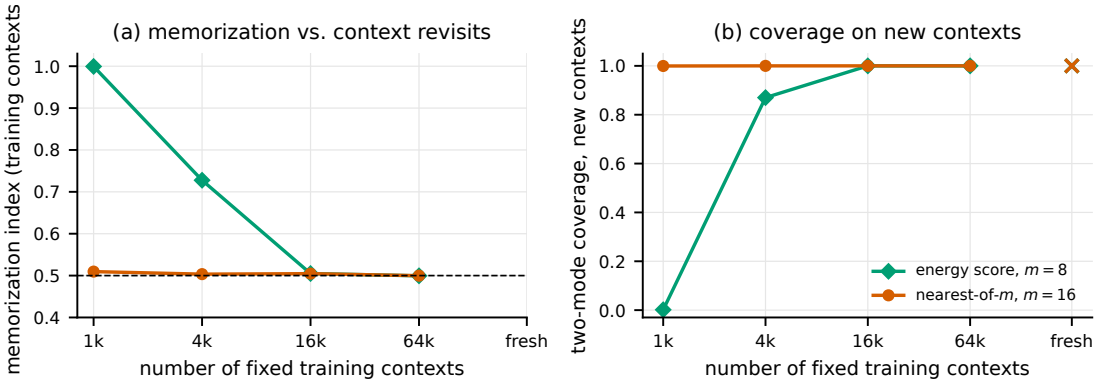


Figure 8: Toy two-branch problem with a nuisance appearance vector (two seeds, m=8 for the energy score and 16 for nearest-of-m). (a) Memorization index on training contexts against the number of fixed training contexts. (b) Two-mode coverage on new contexts; "fresh" draws a new context at every step.

/ 12.6 for FM-2 / FM-50 and 12.2 / 12.3 for IMLE-WM with m = 4 / 16 (seed 0). On the branch-risk task the per-start absolute error of the five-push pit probability is 0.078 (IMLE-WM), 0.086 (FM-50), 0.096 (FM-2) and 0.462 (regression), and the timeout rates are 13.7%, 14.3%, 19.3% and 17% (oracle). Wall-clock for 100 episodes: oracle 100 s, regression 523 s (m_p=1), IMLE-WM 3,633 s, FM-2 9,067 s.

G Negative results

The pilot study tested four additions proposed in our original design; none survived.

Trajectory-level matching. Matching m=4 candidate rollouts of length H=8 as whole trajectories (with a curriculum $H = 1 \to 4 \to 8$) on the push environment was worse than one-step matching at every horizon: best-of-16 position error 1.31 vs. 0.35 px at h=1, 3.89 vs. 2.19 at h=8, 4.67 vs. 3.03 at h=16 (paired bootstrap intervals exclude zero), one-step coverage 0.71 vs. 0.98, and 3.3× the training time; only the error of the sample mean improved (9.18 vs. 9.94 at h=16), the signature of averaging. With two to three modes per step an 8-step trajectory has hundreds of trajectory modes, far beyond m=4, so the matched candidate is the one close to all of them on average. Per-step matching (Section 3) replaced it.

Multi-stage supervision. On the fixed 16×16 token grid, supervising intermediate heads with spatially pooled targets (4×4, 8×8, 16×16) or with PCA-truncated targets (32, 128, 384 components) gave the same coverage as a single head within the bootstrap interval (0.960 and 0.959 vs. 0.967 at K=4) and the same precision (0.916, 0.922 vs. 0.926). The final recipe uses a single head.

Table 17: Fed-back-state ablation on Bouncing Shapes (8M, m=16, 8k steps, two seeds): manifold distance, best-of-16 readout error and spread of the 16 rollouts at h=16 (the true futures spread by 23.6 px), and F1 error. All three predictions recorded before the ablation held (S1–S3 below).

Fed-back state	Manifold	Best-of-16	Spread	F1
One-step matching	2.17	23.2	17.0	8.74
Per-step, feed matched	1.92	**17.7**	26.3	**8.51**
Per-step, feed random candidate	**1.81**	22.9	12.4	8.77
Per-step, feed true frame	2.14	23.6	16.6	8.62

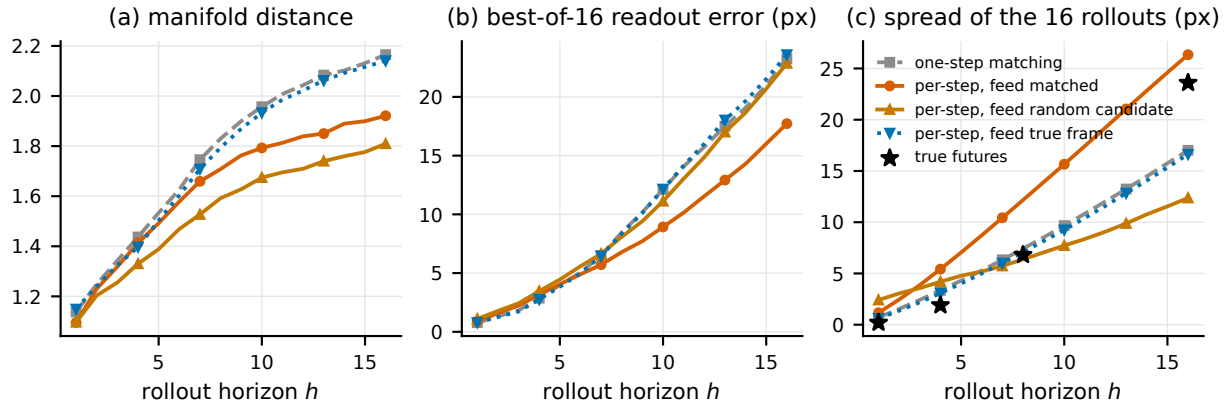


Figure 9: The fed-back-state ablation on Bouncing Shapes (two seeds). (a) Any self-generated context, matched or random, keeps the rollout on the manifold; the true frame does not. (b) Only the candidate nearest the truth improves the best-of-16 error. (c) The matched variant is the only one whose 16 rollouts spread like the true futures at h=16; the random variant jitters at every step but does not form distinct futures, and spreads to half the true spread.

Variance-penalized planning. A CEM cost of mean plus λ times the spread of the imagined goal distances lowered the success rate of the branch-risk task even when the true simulator served as the world model: 78%, 77%, 70% and 50% at $\lambda = 0, 0.5, 1, 2$, with the pit rate rising to 35%. A certain failure has zero variance, so the penalty prefers it to an uncertain success. With IMLE-WM as the model (two seeds, 60 episodes each) the pooled success was 62.5% at $\lambda = 0$ and 61.7% at $\lambda = 0.5$.

Counterfactual action contrast. We had anticipated that the noise z might absorb the effect of the action. It does not: on the push environment the counterfactual precision (the fraction of samples under a random other action that stay away from the true outcome) and the action sensitivity (the change in output per change in action, relative to the true change) are 0.95 and 0.96 with the action on its own FiLM path, 0.92 and 0.97 with the action merged into the noise path, and 0.93 and 0.975 with an added margin term, against 0.83 and 0.94 for regression. There was no shortcut to repair.

Low-dimensional matching and larger budgets on BAIR. See Table 13: m=64 raises the spread-to-error ratio from 0.44–0.51 to 0.53–0.55 at 2.4× the training time, and matching in a 512-dimensional pooled space lowers it to 0.31–0.38; neither approaches FM-50 on distribution metrics.

H Statistical methodology

Per-context intervals are percentile bootstrap intervals (2,000 resamples) over the 400 evaluation contexts of a run, used in the pilot and recipe tables. Seed statistics are means and standard deviations over the

Table 18: Planning with CEM-MPC. Left: stochastic Push-T, 100 episodes × 3 seeds (FM-50: 10 episodes, 1 seed), planning time per MPC step. Right: branch-risk task (push along the pit edge), 100 episodes × 3 seeds, and the model-estimated pit probability after one push on the ice (true 0.33) and after five straight pushes from 200 start states (true 0.43).

Stochastic Push-T	Success	s / step
Regression	$31.7 \pm 3.3\%$	4.3
FM-2	$34.7 \pm 3.1\%$	8.5
FM-50	30%	200
IMLE-WM, m=16	$31.7 \pm 2.1\%$	**4.0**

Branch-risk task	Success	Pit	One-push pit prob. (true 0.33)	Five-push pit prob. (true 0.43)
Oracle (true simulator)	76%	7%	0.33	0.43
Regression	$56.3 \pm 3.4\%$	$24.3 \pm 2.5\%$	0.61 ± 0.10	0.26 ± 0.19
FM-2	$67.0 \pm 1.4\%$	$18.7 \pm 0.9\%$	0.14 ± 0.00	0.38 ± 0.06
FM-50 (estimates only)	–	–	0.31 ± 0.02	0.44
IMLE-WM, m=16	$55.3 \pm 2.6\%$	$31.0 \pm 2.2\%$	0.36 ± 0.03	0.44 ± 0.04

listed seeds. For the Push-T multi-future test set, each model's per-start errors were stored; the start-state interval averages the seeds first and resamples the 128 starts, and the hierarchical interval resamples the seeds with replacement and then the starts, 10,000 times each. Per-step and one-step matching runs with the same seed are paired. All comparisons report the direction and the magnitude of the difference; where an interval includes zero we say so in the text.